INCH-POUND

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DEPARTMENT OF DEFENSE
STANDARD PRACTICE

INSPECTION REQUIREMENTS, DEFINITIONS AND CLASSIFICATION OF DEFECTS FOR PARACHUTES

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1. SCOPE

1.1 Scope. This standard establishes the inspection requirements and a standard classification of manufacturing and final assembly defects for parachutes.

1.2 Purpose. The purpose of this standard is to:

a. Provide a uniform standard of quality for determining the acceptability of parachutes and their components.

b. Standardize the inspection requirements for all parachute manufacturers.

c. Consolidate, into a single document, definitions of all parachute terms.

1.3 Application. This standard should apply when it is referenced in the specification or contract provisions.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in the tables of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this standard, as well as those cited in the tables, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-DTL-9401	Canopy, Parachute, Ring Slot, General Specification for Construction of

(Copies are available online at https://quicksearch.dla.mil)

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document is the issue cited in the solicitation or contract.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D6193 Standard Practice for Stitches and Seams

(Copies of this document are available online at https://www.astm.org/)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this standard takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 General definitions. The following are the definitions for the two classifications of defects used in this standard. Note that some definitions may apply primarily to only round/ballistic type parachutes or only to gliding/ram-air type parachutes or to both.

3.1.1 Major/Critical defect. A defect that may result in failure or reduce the usability of the unit of product for its intended purpose. A major/critical defect is also defined as a dimension out of tolerance, component and or construction deviation which may adversely affect the form, fit or function of the parachute assembly.

3.1.2 Minor defect. A defect that is not likely to reduce the usability of the unit of product for its intended purpose or is a departure from established standards having little bearing on the effective use or the operation of the unit.

3.2 Specific definitions of parachute terms.

3.2.1 Adapter. A rectangular metal fitting with a crossbar incorporated in a parachute harness to permit proper adjustment of harness.

3.2.2 Adapter, harness quick fit. An adapter in which a floating friction grip is used in lieu of a fixed crossbar.

3.2.3 Apex. The center and top most point of an inflated round parachute canopy.

3.2.4 Arm attaching seam. Seams created on cross-cruciform parachute when canopy arms are sewn to canopy center section.

3.2.5 Arm section. Areas of a canopy below the arm attaching seams.

3.2.6 Back pad. A pad attached to the inside of the harness to provide comfort for the wearer and help keep the harness in place.

3.2.7 Back strap. Part of the harness webbing extending across the back of the wearer. It may or may not be adjustable.

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3.2.8 Bag, deployment. A type of container made of fabric and webbings in which a parachute canopy is packed.

3.2.9 Band, lateral. Webbing inserted in the hem of parachute canopies to reinforce edges of fabric and distribute the load. The lower lateral band is inserted in the skirt hem and the upper lateral band is inserted in the vent hem.

3.2.10 Band, pocket. A piece of textile tape or line attached at the outside of the skirt across main seams in such a manner as to cause the gores to be pulled outward at inflation.

3.2.11 Band, reinforcement. A tape or a ribbon inserted in various positions to reinforce weak points in a canopy.

3.2.12 Band, spring, parachute pack opening. A cloth covered steel spring assembly with a hook at each end that rapidly pulls the flaps away from the canopy allowing quick opening of the pack.

3.2.13 Bandstow. A webbing or tape loop sewn to a packtray, deployment bag, or sleeve, for attachment of rubber retainer bands.

3.2.14 Band, retainer. An elastomeric band used to hold folded suspension lines or static lines to deployment bags or parachute packs.

3.2.15 Bias. A cut or seam running obliquely across the yarns of a fabric.

3.2.16 Binding. A piece of tape or fabric folded over and stitched to a raw edge of the fabric to prevent raveling or fraying.

3.2.17 Block stamp wording. Headings for the stamp or markings used to identify a part such as “Contract No:”, “Cage Code.”, etc., which precede the actual identifying information.

3.2.18 Bottom, false. A piece of fabric sewn to the inside of a pack to retain the frame; it also serves as a base for stitching the suspension line retaining loops.

3.2.19 Break cord. A thread or tape, tied between parachute components that is intended to break under desired load during deployment.

3.2.20 Bridle. The cord attaching the pilot chute to the vent of a parachute canopy or to the deployment bag.

3.2.21 Burns, friction. Deterioration of individual yarns caused from heat generated by the rubbing together of a textile and another surface.

3.2.22 Cable, ripcord. A flexible cable, usually made of corrosion resistant steel, joining the locking pins and the ripcord grip.

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3.2.23 Canopy. The portion of a parachute consisting of the drag-producing or lift-producing surface, and the suspension lines extended to one or more mutual confluence points.

3.2.23.1 Canopy, extended skirt. A canopy made from cloth having a flat circular center to which an annular ring is added.

3.2.23.2 Canopy, flat circular. A canopy made from cloth and constructed as a flat circular surface with a center orifice (vent) and consisting of a number of gores stitched together laterally, the joints forming the radial seams.

3.2.23.3 Canopy, guide surface. A canopy made from cloth similar to flat circular canopy except the guide surface panels are attached to the edge of roof panels to enhance stability.

3.2.23.4 Canopy, ribbon. A canopy of flat circular design and composed of concentric cloth ribbons, supported by a number of radial ribbons and smaller supporting tapes.

3.2.23.5 Canopy, ring sail. A canopy of ring slot type developed on a spherical surface by a unique system of gore coordinates, basically shaped as a quarter sphere wherein slots in the gore are crescent shaped rather than trapezoidal (except on the crown).

3.2.23.6 Canopy, ring slot. A canopy of flat circular design made from wide concentric cloth strips with intervening air slots. The number of slots vary depending upon canopy diameter.

3.2.24 Cap seam. The top seam of the pilot chute, which joins the gores and the pilot chute cap (disk or crown) together.

3.2.25 Cascade. The junction of two or more suspension lines before attachment to the risers on gliding type parachutes.

3.2.26 Cell. One or more adjacent air intake passages between two adjacent groups of suspension lines on ram-air or similar gliding type parachutes.

3.2.27 Center section. Top area of a canopy above the arm attaching seams.

3.2.28 Channel canopy. The space or opening through which the suspension lines or radial tapes are passed. It is formed by the overlapping of the fabric in the main seams, or by the addition of cover tape to the drag producing surface, generally on round type parachutes.

3.2.29 Chest protector. A pad fastened to the inside of the harness to absorb shock.

3.2.30 Chordwise. The direction in which the ribs are attached on the upper and lower glide surfaces of gliding type parachutes.

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3.2.31 Chordwise seams. The seams that attach the rib panels to the upper and lower glide surfaces of gliding type parachutes.

3.2.32 Chest strap. A harness strap secured across the chest to prevent the wearer from falling out of the harness.

3.2.33 Chute. The term used interchangeably for the word "parachute".

3.2.34 Chimney effect. A change in the gradual tapering of the gores of the canopy resulting in constriction between the hem and vent.

3.2.35 Clevis. A u-shaped metal fitting with a hole in each end to receive a pin or bolt.

3.2.36 Clip, safety. A special shaped metal fitting used to prevent the accidental opening of the parachute harness release.

3.2.37 Cluster. Two or more parachutes that are attached to a single load and designed to open simultaneously.

3.2.38 Cone, pack. A small cone-shaped metal post, sewn to one of the side flaps of the pack, containing a drill hole through the cone near the top. Grommets positioned on the opposite flap of the pack are placed over the cones and ripcord pins are inserted in the cone hole to keep the pack closed.

3.2.39 Cord, arming. The cord that arms the reefing line cutter or other actuation device when pulled.

3.2.40 Corner, flap. A rectangular tab used on packs to add protection to the canopy when packed.

3.2.41 Cross port vent holes. The vent holes cut in the rib panels of ram-air type gliding parachutes.

3.2.42 Cross seam. The sewn seam between adjacent main seams holding sections of the canopy gore together.

3.2.43 Crown. A fabric panel used to close or cover the apex vent in certain types of parachute canopies. Also may refer to the upper portion of a canopy, particularly a round type parachute.

3.2.44 Cutter, reefing line. A device designed to cut through the reefing line of a canopy, normally incorporating a delay device (mechanical or pyro-technic), a power device (mechanical or pyro-technic), and a knife-edge blade.

3.2.45 Dart. A short tapered seam.

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3.2.46 Diameter, constructed (Dc). The designation of the size of a parachute canopy, based upon design dimensions typically used for round type parachutes.

3.2.47 D-ring. An item of hardware shaped like a "D" into which connector snaps are hooked.

3.2.48 Drogue. Typically a small first-stage parachute canopy in a system used for initial deceleration and stabilization, or extraction.

3.2.49 Elastic, pack-opening. Elastomeric or metal springs with a means of attaching at each end, installed on the pack under tension and used to separate end flaps from side flaps when the parachute ripcord is pulled.

3.2.50 Eye. A small steel-wire loop attached to the pack, into which is fastened a hook of a pack opening elastic.

3.2.51 Flap, bag or pack. A fabric extension on a side or end of the pack designed to enclose and protect the canopy.

3.2.52 Flap, locking pin protector. A flap that covers the locking pins and cones to prevent the pack from being opened by any means other than pulling the ripcord.

3.2.53 Flaring. Method of splitting, taping, and stitching the end of webbing to widen it and prevent it from slipping through an adapter.

3.2.54 Foreign object. Something found in the end item that does not belong there.

3.2.55 Fullness, canopy. Volume added to a gore by puckering and gathering seams of a certain length into small wrinkles or folds along radial tape of a shorter length. Puckering and gathering is evenly spread out along the full length of the seam.

3.2.56 Fullness, seam. Excess cloth between rows of stitching in a seam which is greater than stitch row spacing.

3.2.57 Glide surfaces, upper and lower. The panels forming the top and bottom glide surfaces of a gliding type parachute.

3.2.58 Gore. The portion of a canopy between two adjacent suspension lines or radial seams on a round type parachute.

3.2.59 Handle, ripcord. A loop/handle, generally of metal or rigid material, that provides a grip for pulling manual actuated ripcord cables on ripcord actuated parachutes.

3.2.60 Hardware. All metal fittings used on parachutes, parachute systems, and suspended loads.

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3.2.61 Harness. An arrangement of webbings and hardware designed to conform to the shape of the load in order to secure it properly and to distribute the stress from the opening shock and the weight of the load.

3.2.62 Hem. Fabric folded back upon itself and sewn in this position to form both the peripheral edge and the vent of the canopy.

3.2.63 Hem rigged canopy. A canopy whose suspension lines are attached to the skirt hem and do not pass over the drag-producing surface, generally in reference to round type parachutes.

3.2.64 Hesitator, skirt. A device that restricts the skirt of the canopy from opening, preventing inflation until completion of line stretch, at which time the hesitator line breaks and allows inflation of the canopy.

3.2.65 Holes, drill. Small pinholes used as optional manufacturing aid to indicate location on textile components. Nonpermanent drill holes can be formed using a cold drill. Permanent drill holes resembling a small burn mark can be formed using a hot drill or laser.

3.2.66 Hook,band, parachute container opening. A small formed steel-wire device attached to each end of a pack-opening elastic that hooks into eyes sewn on the pack.

3.2.67 Housing, ripcord. A flexible tube through which the ripcord cable is passed.

3.2.68 Junction. The physical attachment of groups of suspension lines or control lines to each other.

3.2.69 Keeper. A band formed from webbing or elastic. Used to bind excess webbing, secure components to a harness or load, or provide a confluence point for suspension lines and risers.

3.2.70 Knot, clove or half hitch. A type of knot commonly used for attaching the suspension lines of a parachute to the connector links.

3.2.71 Knot, overhand. A simple knot tied in each running end of a piece of cord above a square knot or surgeon's knot to prevent the ends from slipping back through the knot.

3.2.72 Knot, square. A strong knot for joining two cords or lines that does not slip or loosen easily.

3.2.73 Knot, surgeon's. A type of knot typically used to terminate double strand hand tacking, that is similar to a square knot with two turns instead of one for the first knot layer. This type of knot is commonly used in conjunction with a square knot.

3.2.74 Lanyard. A length of cord, webbing, or other material used to retain parachute subassemblies to each other or a device used to activate the deployment of a parachute system.

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3.2.75 Leading edge. The front area in which the air enters the canopy, generally for gliding type parachutes but sometimes used for round type parachutes. May also refer to edge facing upstream into the airflow of a ribbon or cloth section for ribbon and ring slot parachutes.

3.2.76 Leading edge seams. The two seams sewn on the leading edge in the spanwise direction on gliding type parachutes.

3.2.77 Leg strap. The retention strap, which is part of the harness, used to secure the harness to the wearer's legs.

3.2.78 Lift web. The main harness webbing which runs vertically along the front torso or payload.

3.2.79 Line continuity. Free and clear routing to certain areas on canopy to ensure proper steering of canopy.

3.2.80 Line, control. Used to steer and maneuver canopy, also known as steering or brake lines.

3.2.81 Line, guide or control. Two or more parachute lines that run from a slot or orifice in a steerable canopy to the risers. Control lines for a gliding parachute are connected from the trailing edge to a cascade and routed to the risers.

3.2.82 Line, reefing. A length of cord or line passed through rings or loops on the skirt of the drag-producing surface to delay or control opening of the canopy.

3.2.83 Line, spreader. X pattern webbing used in the LCADS HV as a deployment aid to ensure that the suspension lines maintain continuity.

3.2.84 Line, static. A line, webbing, or cable used to open a pack or to deploy the canopy, one end of which is fastened to the pack, canopy, or deployment bag, and the other to some part of the launching vehicle.

3.2.85 Line, suspension. Cords or webbing that connect the drag-producing surface to the harness or the load.

3.2.86 Line, suspension, flat. A line made of tape or webbing.

3.2.87 Lines, vent. The lines that cross the vent of a canopy.

3.2.88 Line reinforcement (V tabs or butterfly). A tape wrapped tightly about the suspension line and sewn to the suspension line and skirt hem.

3.2.89 Line group, suspension. All suspension lines on one connector link.

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3.2.90 Line set, suspension. All suspension lines on one parachute assembly.

3.2.91 Link, connector. An item of hardware used to connect suspension lines at the load attaching point.

3.2.92 Link, connector, separable. A readily separated link to facilitate assembly of risers to suspension lines.

3.2.93 Loop, locking. A loop sewn to the deployment bag or canopy to sequence the opening of a parachute assembly.

3.2.94 Loop, retaining. A loop of webbing or tape used to hold folded lines or excess webbing in position.

3.2.95 Loop, stow. A tape, webbing, or elastic band on a pack or deployment bag that holds suspension lines in position during the packing and opening process.

3.2.96 Lower cascade. The junction of control lines closest to the risers on gliding type parachutes.

3.2.97 Lug. A flat metal fitting attached to the ends of the harness webbing to provide attachment to the harness release.

3.2.98 Main seam. The seam sewn from skirt to vent/apex holding gores together and used as a guide for the suspension line or radial tape on a round type parachute.

3.2.99 Nominal tension. One percent of the minimum breaking strength of the main strength-bearing material. A maximum pull for nominal tension should be 30 pounds. This definition applies when MIL-DTL-9401 is referenced.

3.2.100 Pack or container. The container that encloses the canopy or deployment bag in a packed condition and which provides a means of opening to allow deployment of the canopy.

3.2.101 Pack cover. A piece of duck or canvas material with a static line attached used to cover a packed canopy.

3.2.102 Pack frame. A rigid or flexible frame used to maintain the shape of the pack.

3.2.103 Panel. Individual section of fabric sewn to another fabric section, using cross seams to form a gore. Also refers to the continuous length of cloth cut from a pattern.

3.2.104 Panel section. Area of a panel that is subdivided by seams and/or reinforcement tape.

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3.2.105 Parachute. An assembly consisting of canopy, risers, bridles, deployment bag, and in some cases, a pilot chute. Pack and attaching webbings (harnesses) are a part of the parachute when they are built into the suspended load as an integral part of the load.

3.2.105.1 Parachute, airdrop (cargo chute). A parachute designed to deliver equipment or supplies from aircraft in flight.

3.2.105.2 Parachute, approach, landing. A parachute used in flight to improve jet aircraft flight characteristics during normal landing approach or in approach under marginal weather conditions.

3.2.105.3 Parachute, back-type. A parachute designed for attachment to the wearer's back.

3.2.105.4 Parachute, chest-type. A parachute designed for attachment to the wearer's chest.

3.2.105.5 Parachute, deceleration (drag). A parachute used on jet aircraft to decrease landing roll.

3.2.105.6 Parachute, extraction. A parachute used to extract cargo from aircraft in flight and to deploy cargo parachutes.

3.2.105.7 Parachute, gliding. A parachute specifically designed to glide, generally of the ram-air type configured as a rectangular or elliptical wing, sometimes referred to as a square parachute.

3.2.105.8 Parachute, personnel. A parachute used to lower personnel from aircraft in flight.

3.2.105.9 Parachute, recovery system. Normally these systems are comprised of static lines, a pilot chute, deployment bag(s), first stage parachute (used to decelerate and stabilize the load), intermediate parachute (used to further decelerate the load), final recovery parachute (used for final delivery of load to earth), and controlling/actuating devices.

3.2.105.10 Parachute, reserve. A secondary parachute, worn while making a premeditated parachute jump, to be used in the event that the main parachute fails.

3.2.105.11 Parachute, round. A parachute in a generally round configuration that has no or very minimal glide capability and is generally considered to have a ballistic trajectory. May refer to a number of round configurations or to any non-gliding configuration.

3.2.105.12 Parachute, seat-type. A parachute designed for attachment to the saddle of the harness and may serve as a seat cushion in certain types of aircraft.

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3.2.105.13 Parachute, troop. A parachute used by paratroopers for a premeditated jump over a designated area.

3.2.106 Pilot chute. A small parachute used to accelerate deployment of the main parachute. Some pilot chutes are equipped with a metal spring device to open the canopy when released from the pack.

3.2.107 Pilot chute frame. A wire frame or spring used in the pilot chute to initiate the opening action of the chute upon release from the pack.

3.2.108 Pin, locking (ripcord). A short metal pin attached to the ripcord cable that is inserted into a pack cone or other device to secure the flaps as a function of closing a parachute pack.

3.2.109 Pocket, log record. A small patch pocket sewn to the pack or riser for carrying the parachute log record.

3.2.110 Radial seam. The seam sewn from skirt to vent holding gores together and used as a guide for the suspension lines or radial tapes for round type parachutes. This term is normally used in lieu of main seams for ribbon-type parachutes.

3.2.111 Reefing rings. The rings sewn to the skirt of a canopy at suspension line attaching points through which the reefing line is passed.

3.2.112 Reinforcement tapes. Any tape or webbing sewn to a pack or canopy at a point of high load to distribute the load and strengthen the fabric.

3.2.113 Release, canopy. A device that is designed to permit rapid separation of the canopy and risers from the suspended load.

3.2.114 Release, harness. A device that is designed to permit rapid release of the harness from the wearer.

3.2.115 Release, 3-ring. The rings and webbing that compose a riser/canopy release mechanism consisting of three interlocked rings of graduated size.

3.2.116 Ribs. The airfoil shaped panels that are attached to the upper and lower gliding surfaces in the chordwise direction, generally for a ram-air gliding type parachute but may refer to internal structure of some round type parachutes.

3.2.117 Rib panel reinforcements. The tapes that are sewn on the ribs.

3.2.118 Rib pocket. A pocket made by sewing lengths of tape to the pilot chute to contain the ends of the frame.

3.2.119 Ripcord. A device that secures the folded parachute in the pack and provides a release for the canopy in deployment.

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3.2.120 Riser. The portion of a suspension system between the lower end of a group of canopy suspension lines, and the point of attachment to the load.

3.2.121 Saddle. The part of the main lift web of the harness that provides a seat or sling for the wearer.

3.2.122 Searing. A method of cutting or sealing ends of heat sensitive material, such as nylon, using a hot tool to prevent raveling.

3.2.123 Section. One of the pieces of cloth making up the gore of a round type canopy.

3.2.124 Serving. A method of wrapping or binding the ends of lines or cords to prevent raveling.

3.2.125 Shoulder strap. The harness webbing that crosses the shoulders of the wearer.

3.2.126 Skirt. The reinforced hem forming the edge of the canopy.

3.2.127 Sleeve, deployment. A deployment aid using fabric and webbing which may enclose the full length of the canopy.

3.2.128 Slider. A reefing device that controls the opening of a canopy by sliding down the suspension lines during canopy inflation.

3.2.129 Slider reinforcements. The webbing and tapes that are sewn on the slider.

3.2.130 Spanwise. The direction of the leading and trailing edges, perpendicular to the direction of flight on gliding type parachutes.

3.2.131 Spanwise seams. The seams joining the spanwise panels of the upper and lower gliding surfaces of some gliding type parachutes.

3.2.132 Spring assembly, ejector. A device that assists the deployment of the pilot chute.

3.2.133 Trailing edge. The rear edge of the canopy to which the control lines are attached on a gliding type parachute. May also refer to the edge down-stream of the airflow on a ribbon or section of cloth for ribbon and ring slot parachutes.

3.2.134 Trailing edge seams. The seams sewn on the trailing edge in the spanwise direction on gliding type parachutes.

3.2.135 Upper cascade. Any junction of control lines above the lower cascade.

3.2.136 Vent. The opening at the top of a parachute canopy, generally referring to round type parachutes.

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3.3 Specific definitions of parachute hardware terms and conditions.

3.3.1 Bend test. A prescribed method used to test ripcord pins for rigidity.

3.3.2 Brazing. A joining process using a high temperature filler metal to bond adjoining metals.

3.3.3 Composite. Material made from two or more components with different properties that when combined, produce a material with characteristics different from the individual components.

3.3.4 Crack. A clear crystalline break caused by localized stresses exceeding the rupture strength.

3.3.5 Eyelet. Small metal reinforcement for a hole in fabrics. It is thinner and smaller than a grommet and has no washer.

3.3.6 Grommet. A metal eyelet that interlocks with a washer, used as reinforcement around a hole in fabric.

3.3.6.1 Grommet, plain. A metal eyelet that interlocks with a plain washer.

3.3.6.2 Grommet, rolled rim. Interlocks with spur washers to provide a firm, tight setting.

3.3.7 Hardness. Resistance of metal to plastic deformation usually from indentation or penetration by means such as abrasion, drilling, impact, scratching and/or wear. Results can be used to determine tensile strength in some materials, typically metal.

3.3.8 Laps. Folds in hot metal that have been rolled or forged into the surface but not welded to it.

3.3.9 Seam. An opening on the surface of metal that has been closed but not welded.

3.3.10 Soldering. A joining process using a low temperature filler metal to bond adjoining metals.

3.3.11 Swaging. The joining of metal parts by pressure, such as in attaching ripcord locking pins to ripcord cable.

3.3.12 Washer. A thin plate with a hole that works with a grommet to make a secure setting.

3.3.12.1 Washer, plain. Basic reinforcement which provides support for the grommet setting.

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3.3.12.2 Washer, spur. A washer that has interlocking teeth which provide a firm, tight grasp for the grommet setting.

3.3.13 Welding. A joining process that melts similar adjoining metals to form a bond.

3.4 Specific definitions of parachute sewing terms and conditions.

3.4.1 Backstitch or backtack. The start and/or finish of a row of stitching by sewing back over the original stitching a short distance to retard raveling.

3.4.2 Bartack. A concentrated row of stitching used to reinforce points of stress. Often resembles a zigzag pattern.

3.4.3 Basting. A temporary stitching, usually with long loose stitches.

3.4.4 Bird nest (stitching nest). Tangled mass of thread resembling a bird's nest, which can occur on the needle or bobbin side.

3.4.5 Broken stitch. A break in sewing thread.

3.4.6 Bunched stitching. Non-uniform distribution of stitches resulting in excessive stitching in a single area.

3.4.7 Cross box / box-x. A sewing pattern that consists of a stitched box with an X sewn from corner to corner.

3.4.8 Crowd stitching. The intentional concentration of a stitch pattern to secure the end of a stitch pattern, usually at the end of a row of stitching to reinforce the stitch.

3.4.9 Double W / 4-point. A sewing pattern that consists of two parallel rows with two or more equally spaced crossing rows in between them forming a double W (or 3-point), or a 4-point. Also referred to as a cross stitch.

3.4.10 Folder. A device used as an attachment to a sewing machine to guide and fold fabric.

3.4.11 Gauge. The measured distance between needles on a sewing machine. Also referred to as the capacity of a folder.

3.4.12 Hand tack. A stitch produced by hand sewing.

3.4.13 Kink. A short length of thread that protrudes from the surface of a stitch line that doubles back on itself to form a twisted or knotted loop. Also referred to as knotty thread.

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3.4.14 Loop. A continuous thread that protrudes from the surface of a stitch line, forming a raised loop instead of a flat stitch.

3.4.15 Loose stitches. Thread that does not lie smoothly on the surface of the fabric.

3.4.16 Margin. The space from the outer row of stitching to edge of the fabric or fold of the fabric.

3.4.17 Missing stitches. Space between stitches in the same row in which there is no thread.

3.4.18 Needle damage. The partial or complete yarn severance or fiber fusing caused by the needle passing through a fabric during sewing.

3.4.19 Open seam. One or more consecutive broken stitches or two or more consecutive skipped stitches in a seam.

3.4.20 Overedge. The stitching around the outer edges of fabric that prevents the edges from raveling or fraying.

3.4.21 Overfold. An excess of material causing the edge of an inner fold to double.

3.4.22 Overlap. The distance one material, thread, or cord extends over another material, thread, or cord.

3.4.23 Pleat. A fold sewn in fabric. Pleats may be inserted by design or may be the result of unintentional gathering of cloth.

3.4.24 Ravel. To come undone, separate, untwist, unwind, or unweave, leaving a frayed or ragged edge.

3.4.25 Replacement stitching. Removal of defective stitching followed by resewing the seam or stitch pattern.

3.4.26 Runoff. Stitching that run off the intended seam, fabric, tape, webbing, or cord.

3.4.27 Skipped stitches. Long float in a stitch line due to stitches that fail to interlace or interloop.

3.4.28 Stitch per inch. Frequently referred to as SPI. In straight stitching, the number of needle penetrations per linear inch when the threads are interlocked. Zig-zag stitching may be defined either by number of points along one side or by total stich count.

3.4.29 Stitch reinforcement. Stitch repair which lies adjacent to or superimposes defective stitching. Defective stitching is not removed.

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3.4.30 Stitch repair. A section of stitching that is removed and replaced, or reinforced, due to defective stitching, characterized by stitch overlap on each end.

3.4.31 Stitch splicing. Restarting a stitch row that has ceased because, for example, a bobbin ran out of thread. Stitch splice is characterized by stitch overlap on one end.

3.4.32 Stitching. Sewing, normally by machine, to join two or more pieces of fabric by thread.

3.4.33 Tight stitches. Thread under excess tension causing the interlocking knots to lie on either surface of fabric, or causing puckering of fabric.

3.4.34 Uneven stitching. Inconsistent stitch length within a measured distance.

3.4.35 Tucks. A shortening of material caused by pulling fabric up in folds and stitching across the gathered fabric, generally parallel to the seam.

3.4.36 Underfold. Insufficient material folded inside a seam.

3.4.37 Zigzag stitch. A stitch formation of alternating left and right throw stitches.

4. GENERAL REQUIREMENTS

4.1 Limitation of use. The conditions described in the defect listing herein will not be used for manufacturing purposes to extend specified tolerances of drawing requirements. Items are required to meet drawing requirements; therefore, immediate action will be taken to correct the causes of all defects. This document is not intended to cover all defects that may be encountered during the manufacture of parachutes and components. When deviations from the drawings or specifications are encountered that are not covered by this standard, final disposition by the procuring activity shall be required.

4.2 Interpretation of classification of defect tables. Conditions listed in tables III through XXI apply only in the direction of tolerance specified in the applicable drawings, specification, or standard. These conditions are defects that fall outside of the applicable drawing, specification, or standard requirements. When the tolerance is called out in one direction, for example, minus, the variation shall apply only in the minus direction. When the tolerance is called out plus or minus, the variation may be taken in either direction. Notes are included as necessary under description of defects to ensure clarity and preclude the misinterpretation of requirements.

5. DETAILED REQUIREMENTS

5.1 Inspections and tests.

5.1.1 Component and material inspection. Components and materials shall be inspected as specified in the applicable end item specification.

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5.1.2 In-process inspection of the product.

5.1.2.1 Material cutting. Controls shall be established to assure the accuracy and serviceability of patterns and the accuracy of webbing and cord cutting tables.

5.1.2.2 Manufacturing. Inspection will be performed at the inspection stations established by the contractor to control the quality of the parachute systems and components during manufacture. Any defects found during in-process inspection are to be repaired in accordance with applicable manufacturing specification.

5.1.3 Final inspection of the product.

5.1.3.1 Inspection of items. Each parachute assembly or component thereof will be inspected for all manufacturing operations and material requirements that have not been previously inspected and accepted through in-process inspection. The incorrect assembly or construction of any components or individual parachute system will be cause for rejection. The acceptance number for major/critical defects shall be zero and the number of minor defects shall not exceed the quantities set forth in table I or II.

5.1.3.2 Inspection of cloth, tape, and webbing. The cloth, tape, and webbing shall be examined visually to detect holes, snags, strains, stains, or other defects. Unless otherwise specified herein, encountered defects shall be classified in accordance with the classification of defect tables in the applicable material specification. Any material defect in the finished canopy or component (regardless of cause) that are minor will be acceptable, and any major or critical defect will be cause for the canopy or component to be rejected.

5.1.3.3 Inspection of seams, hems, and stitching. The seams of canopies and pilot chutes will be examined for conformance with drawing details by projection of a 40-watt (minimum) fluorescent light through the canopy. Canopies made from tape and webbing shall be examined visually. Seams and hems shall be examined for the proper number of stitches per inch, broken or missing stitches, and proper construction. Defects discovered during these examinations will be classified in accordance with the classification of defect tables herein.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This document is intended for defining the major/critical and minor classifications necessary to determine the acceptability of various parachutes for Government procurement purposes.

6.2 Subject term (key word) listing.

Canopy
Cargo

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Cascade
Chute
Deployment
Harness
Hems
JPADS
Main
MC-4
MC-6
MIPRS
Pack
Personnel
RA-1
Reserve
Seams
SF10-A
Skirt
Sleeve
Stitching
T-11
T-11R
Tape
Webbing

6.3 Changes from previous issue. The margins of this standard are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE I. Personnel parachute systems - guideline for maximum allowable errors per item.

No.	Item	Classification Major/Critical	Minor
1	Canopies (solid cloth, round, extended skirt)		
	Troop and rescue	0	8
	Emergency 24 feet	0	5
	Emergency 26 feet	0	6
	Emergency 28 feet	0	6
2	Canopies (square, gliding type, main and reserve)	0	8
3	Canopies (cross-cruciform)	0	8
4	Pilot chute and drogue	0	3
5	Deployment bag, diaper, and sleeve	0	3
6	Pack/container		
	Troop and rescue (excluding static line)	0	4
	Emergency (except chest style)	0	5
	Chest style (including reserve)	0	3
	Container (integrated main and reserve) without harness, MC-4 and RA-1	0	5
	Container (integrated main and reserve) with harness, MC-4	0	9
7	Integrated pack and harness	0	8
8	Harness	0	5
9	Risers	0	1
10	Static lines, bridle lines, straps, lanyards, pockets, and cloth ripcord handles	0	2
11	Spring assembly, ejector	0	3

NOTE: In all of the following tables, all references in the form of "up to" are inclusive

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TABLE II. Cargo, deceleration, extraction, recovery, and special weapon systems - guideline for maximum allowable errors per item.

No.	Item	Classification Major/Critical	Minor
1	Canopies (solid cloth, round, extended skirt)		
	Up to 25 feet in diameter	0	10
	Greater than 25 feet in diameter	0	20
	Greater than 50 feet in diameter	0	30
	Greater than 75 feet in diameter	0	40
2	Canopies (ribbon and ring slot)		
	Up to 25 feet in diameter	0	10
	Greater than 25 feet in diameter	0	20
3	Canopies (glide surface)	0	5
4	Canopies (square, gliding type)		
	Up to 500 sq. ft.	0	10
	Greater than 500 sq. ft. and up to 2000 sq. ft. (JPADS 2K, 2K1T, 4K)	0	30
	Greater than 2000 sq. ft. (JPADS 10K)	0	50
5	Pilot chute and drogue	0	3
6	Deployment bags and packs	0	10
7	Risers, static lines, bridles, straps, retainers, and adapter webs	0	5
8	Soft link	0	1

NOTE: In all of the following tables, all references in the form of "up to" are inclusive

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Table III. Classification of defects - canopies: solid and extended skirt parachutes - personnel.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Vent reinforcement band overlap		
	1/4 inch or less, minus		201
	Greater than 1/4 inch, minus	107	
	Greater than 1 inch, plus		202
8.	Vent hem outside stitching margin from edge of webbing		
	1/16 inch up to 3/16 inch		203
	Greater than 3/16 inch	108	
	All rows of stitching not through reinforcement webbing	109	
9.	Vent loop length (finished dimension)		
	1/2 inch or less, plus		204
	Greater than 1/2 inch, plus	110	
	Minus direction any length	111	
10.	Vent line length (finished dimension)		
	1/4 inch or less		205
	Greater than 1/4 inch	112	
11.	Variation in vent line length per set		
	1 percent or less		206
	Greater than 1 percent	113	
12.	Twisted vent line (flat)	114	
13.	Twisted vent line (core or coreless braid)		
	Greater than 360 degrees		207
	Greater than 360 degrees, causing kinks in line when relaxed	115	
14.	Gore width at vent, centerline to centerline on main seam or as defined in TDP		
	1/4 inch or less		208
	Greater than 1/4 inch	116	

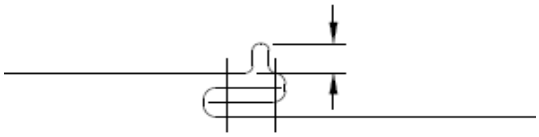
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Table III. Classification of defects - canopies: solid and extended skirt parachutes - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
15.	Gore width at skirt, centerline to centerline on main seam or as defined in TDP		
	1/2 inch or less		209
	Greater than 1/2 inch	117	
16.	Underfold (raw edge exposed in main or cross seam)	118	
17.	Overfold in main seam		
	1/2 inch or less, or extended into center channel		210
	Greater than 1/2 inch	119	
18.	Underfold in main seam		
	A 4-needle seam not caught by three rows of stitches by at least 1/16 inch	120	
	Three intermittent underfolds, up to half the seam width of a 2-needle seam, 2 inches long or less, spaced closer than 12 inches apart		211
	More than three intermittent underfolds up to half the seam width, 2 inches long or less, spaced closer than 12 inches apart	121	
	Underfold, up to half the seam width, greater than 4 inches in length	122	
	More than half the seam width, any length	123	
19.	Overfold and underfold in cross seam		
	Underfold, 1/8 inch to 3/16 inch from folded edge of seam, 8 inches long or less		212
	Underfold, greater than 8 inches long	124	
	Up to three intermittent underfolds, 3/16 inch from folded edge of seam, 2 inches long or less, spaced further than 12 inches apart		213
	More than three intermittent underfolds, 3/16 inch from folded edge of seam, 2 inches long or less, spaced closer than 12 inches apart	125	
	Overfold, quarter to half of seam width, full length of seam		214
	Overfold, greater than half of seam width, any length	126	
20.	Overfold and underfold in SF10-A and T-11R main seam and cross seam		
	Overfold		
	1/8 inch up to 1/4 inch, any length		215
	Over 1/4 inch, any length	127	
	Underfold		
	1/8 inch up to 1/4 inch, any length		216
	Over 1/4 inch, any length	128	
	1/8 inch up to 1/4 inch, greater than 4 inches in length per seam	129	
	More than three intermittent underfolds, 1/8 inch up to 1/4 inch per seam, 2 inches long or less, spaced closer than 12 inches apart	130	
21.	Fullness in seam width, as measured per the following diagram:		

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Table III. Classification of defects - canopies: solid and extended skirt parachutes - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
			
	Main seam		
	1/8 inch up to 1/2 inch		217
	Greater than 1/2 inch	131	
	Cross seam		
	1/8 inch up to 1/4 inch		218
	Greater than 1/4 inch	132	
	Main seam and cross seam, SF10-A and T-11R	133	
	Greater than 1/8 inch		
22.	Puckering and gathering sewn in to create canopy fullness not evenly spaced along radial tape of SF10-A canopy, or not present along the full length of seam		219
23.	Wrinkles and pleats in cross and main seams		
	Cross seams		
	1/16 inch up to 1/8 inch and one per seam		220
	1/16 inch up to 1/8 inch and more than one per seam	134	
	1/16 inch up to 1/8 inch and more than six per canopy	135	
	Greater than 1/8 inch	136	
	Main seams		
	1/16 inch up to 1/8 inch and one per seam		221
	1/16 inch up to 1/8 inch and more than one per seam	137	
	1/16 inch up to 1/8 inch and more than six per canopy	138	
	Greater than 1/8 inch	139	
24.	Vent and skirt hems improperly formed (four rows of stitching)		
	Raw edge showing	140	
	Intermittent underfold where edge is caught with two rows of stitching		222
	Underfold where edge is caught with one row of stitching	141	
25.	Vent and skirt hems improperly formed (two rows of stitching)		
	Raw edge showing	142	
	Intermittent underfold, half of seam width or greater, where edge is caught with one row of stitching	143	
26.	Matching of centerlines of seams when vent and skirt hems are folded		
	1/8 inch up to 1/4 inch		223
	Greater than 1/4 inch	144	
27.	Wrinkles or pleats in skirt and vent hems that do not extend into the gore fabric		
	1/4 inch up to 1/2 inch		224
	Greater than 1/2 inch and more than one per gore	145	

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Table III. Classification of defects - canopies: solid and extended skirt parachutes - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
	Greater than 1/2 inch and more than twelve per canopy up to 35 feet diameter	146	
	Greater than 1/2 inch and more than fourteen per canopy greater than 35 up to 65 feet diameter	147	
	Greater than 1/2 inch and more than eighteen per canopy greater than 65 feet diameter	148	
28.	Wrinkles or pleats in skirt and vent hems extending into the gore fabric		
	1/16 inch up to 1/8 inch long and one per gore		225
	1/16 inch or greater, and more than one per gore	149	
	1/16 inch or greater, and more than six per canopy	150	
	Greater than 1/8 inch long	151	
29.	Skirt hem outside stitching margin from edge of webbing		
	1/16 inch up to 3/16 inch		226
	Greater than 3/16 inch	152	
	Any row of stitching not through skirt band webbing	153	
30.	Skirt hem formed by more than two pieces of tape or webbing, unless otherwise specified on the applicable drawing	154	
31.	Tape or webbing used to form skirt hem does not extend across a minimum of four gores	155	
32.	Tucks		
	1/4 inch wide or less, and 1/2 inch long or less		227
	Greater than 1/4 inch wide regardless of length	156	
	Greater than 1/2 inch long regardless of width	157	
33.	Mismatching of cross seams		
	First seam from vent, 2 inches or greater and 2-1/2 inches or less		228
	First seam from vent, greater than 2-1/2 inches	158	
	All other seams, greater than 2-1/2 inches and 3 inches or less		229
	All other seams, greater than 3 inches	159	
	All seams on SF10-A, greater than required by TDP	160	
	T-11R cross seam misalignments greater than required by TDP	161	
34.	Vent lines out of sequence	162	
35.	Suspension and vent line overlap on canopy		
	1/8 inch or less, minus		230
	Greater than 1/8 inch, minus	163	
	Greater than 1/4 inch, plus		231
36.	Suspension lines not in proper channel or lines crossed	164	

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Table III. Classification of defects - canopies: solid and extended skirt parachutes - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
37.	Cut, frayed, or damaged suspension, control or vent lines	165	
38.	Variation in suspension line length per line group, between vent and skirt		
	1/2 percent or less		232
	Greater than 1/2 percent	166	
39.	Variation in suspension line length per set, between vent and skirt		
	1 percent or less		233
	Greater than 1 percent	167	
40.	Variation in suspension line length per line group, between skirt and attaching point NOTE: Does not apply to SF10-A or T-11R.		
	1/2 percent or less		234
	Greater than 1/2 percent	168	
41.	Variation in suspension line length per set, between skirt and attaching point NOTE: Does not apply to SF10-A or T-11R.		
	1 percent or less		235
	Greater than 1 percent	169	
42.	Suspension line length, main seams, extended gore trailing edge length or break slots out of symmetry on SF10-A or T-11R canopy	170	
43.	Twisted suspension line, between vent and skirt (flat)		
	In channel, 360 degrees or less		236
	In channel, greater than 360 degrees	171	
44.	Twisted suspension line between vent and skirt (core or coreless braid)		
	Greater than 180 degrees up to 360 degrees		237
	Greater than 360 degrees	172	
45.	Twist between line reinforcement at skirt and 3 inch zigzag stitching above the skirt, when suspension lines are continuous through canopy (measured by pulling line under 5 pounds of tension between vent and line reinforcement and observing the rotation of zigzag stitching)		
	Greater than 90 degrees up to 180 degrees		238
	Greater than 180 degrees	173	
	Line not placed between two ply's of material	174	
46.	Twisted suspension line (flat) between skirt and attaching point, except hem rigged canopy	175	
47.	Twisted suspension line (flat), hem rigged canopy		
	Greater than 180 degrees up to 360 degrees		239
	Greater than 360 degrees	176	

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Table III. Classification of defects - canopies: solid and extended skirt parachutes - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
48.	Twisted suspension line between skirt and attaching point (core and coreless braid)		
	Greater than 360 degrees		240
	Greater than 360 degrees causing kinks in the line	177	
49.	Twisted suspension line between suspension line loop and attaching point on SF10-A canopy		
	Greater than 180 degrees	178	
50.	Twisted suspension line between skirt and connector link		
	Greater than 90 degrees up to 180 degrees		241
	Greater than 180 degrees	179	
51.	Open seams	180	
52.	Stitching		
	Stitch per inch defects size 3 - 6 thread:		
	One stitch per inch more or less than the number specified, up to 2 inches in length, and one place per seam		242
	More than two stitches per inch more or less than the number specified	181	
	More than one stitch per inch more or less than the number specified, longer than 2 inches in length	182	
	More than one stitch per inch defect per seam	183	
	Stitch per inch defects size B - FF thread:		
	Two stitches per inch more or less than the number specified, up to 2 inches in length, and one place per seam		243
	Two stitches per inch more or less than the number specified, longer than 2 inches in length	184	
	More than two stitches per inch more or less than the number specified	185	
	More than one stitch per inch defect per seam	186	
	Additional stitch per inch defects, size B and E thread, on ASTM D6193 Type 301 or 401 stitch		
	More than fourteen stitches per inch	187	
	Fourteen stitches per inch, longer than 2 inches in length	188	
	Fourteen stitches per inch, more than one location per seam, spaced closer than 12 inches apart	189	
	Stitch per inch defects in localized areas of reinforcement or excess thickness classified in this table as "major defect"		244
	NOTE: Stitch per inch defects classified in this table as "minor defect", in localized areas of reinforcement or excess thickness, shall not be classified as a defect.		
	Stitch pattern defects:		
	Any broken or missing stitch	190	
	Two or more consecutive skipped stitches	191	
	Missing or incomplete stitch pattern, for example 3-point W, box-x, bartack, zigzag	192	

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Table III. Classification of defects - canopies: solid and extended skirt parachutes - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Backstitch missing	193	
	Backstitch short		
	Greater than 1/16 inch up to 1/8 inch		245
	Greater than 1/8 inch	194	
	Backstitch long, greater than 1/4 inch		246
	Runoff, two or more stitches onto a single cloth, or beyond seam or hem margins		
	1/4 inch or less		247
	Greater than 1/4 inch	195	
	Zigzag stitching where less than 85 percent of the stitches fully penetrate the suspension line	196	
	Stitch tension too loose or too tight		
	3 inches long or less, one per seam		248
	Greater than 3 inches long per seam	197	
	More than one location per seam	198	
	Stitch pattern dimensions not as specified, stitch patterns less than 2 inches in length		
	Greater than 1/16 inch up to 1/8 inch		249
	Greater than 1/8 inch	199	
	Stitch pattern dimensions not as specified, stitch patterns 2 to 5 inches in length		
	Greater than 1/8 inch, minus	101a	
	Greater than 1/8 inch, plus		250
	Stitch pattern dimensions not as specified, stitch patterns greater than 5 inches in length		
	Greater than 1/8 inch up to 1/4 inch		251
	Greater than 1/4 inch, minus	102a	
	Greater than 1/4 inch, plus		252
	Kinks, intermittent, spaced closer than 2 inches apart		253
	Birds nest (stitching nest), to include masses of knotted and tangled thread	103a	
	Repairs to stitching when repairs are forbidden by end item TDP	104a	
	Loose or raveled bartack stitching	105a	
	Repairs to bartack stitching other than complete removal and replacement	106a	
	Bartack stitch pattern length not as specified	107a	
	Box-x stitch pattern length not as specified	108a	
53.	Stitch overlaps on SF10-A and T-11R main seams with seam lengths 240 inches or less		
	More than five overlaps per main seam	109a	
	Repairs on more than half the number of main seams on the canopy	110a	
	Repairs spaced closer than 36 inches apart per main seam	111a	
54.	Frequency of stitch overlaps on canopy (from stitch repairs or splices) that are not otherwise specified in this defect table, shall be classified according to the following table		
	More than the number of overlaps in the table	112a	
	NOTE: One stitch repair or splice will result in two overlaps, one overlap on either end of the original stitch, regardless of number of needles used (single		

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Table III. Classification of defects - canopies: solid and extended skirt parachutes - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
	or multi-needle machine).		
	Length of seams (inches) / Overlaps		
	up to 2 — 0		
	greater than 2 up to 12 — 2		
	greater than 12 up to 72 — 4		
	greater than 72 up to 240 — 6		
	greater than 240 — seam length (in) multiplied by .025		
	Skirt band — seam length (in) multiplied by .02		
55.	Length of stitch overlaps		
	Overlap less than 2 inches	113a	
	Overlap 4 inches up to 7 inches		254
	Overlap greater than 7 inches	114a	
56.	Damage to textiles, not including cuts, tears and holes		
	1/8 inch long or less and one per canopy up to 35 feet diameter		255
	Greater than 1/8 inch long or more than one per canopy up to 35 feet diameter	115a	
	1/8 inch long or less and two per canopy greater than 35 feet diameter		256
	Greater than 1/8 inch long or more than two per canopy greater than 35 feet diameter	116a	
57.	Cuts, tears, holes (except necessary drill holes) to canopy cloth		
	Damage resulting in cuts, tears, or holes involving three or more warp or filling yarns	117a	
58.	Cuts, tears, darns and repairs on mesh panels	118a	
59.	Cuts, tears, holes, darns and repairs to tape or webbing	119a	
60.	Darns to canopy cloth		
	Darns, more than one per section	120a	
	Darns, more than three per canopy	121a	
	Any darn on T11-R canopy	122a	
61.	Splicing of cord, webbing or tape, except hem tape, unless otherwise specified	123a	
62.	Improper routing of control line grouping	124a	
63.	Pocket band free length		
	1/8 inch or less		257
	Greater than 1/8 inch	125a	
64.	Pocket depth, centerline main seam to stitch pattern on skirt band		
	1/8 inch or less		258

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Table III. Classification of defects - canopies: solid and extended skirt parachutes - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
	greater than 1/8 inch	126a	
65.	Suspension line attaching loop, up to 2 inches in length		
	NOTE: On SF10-A this refers to the physical loop at bottom of line attachment loop.		
	1/8 inch or less		259
	Greater than 1/8 inch	127a	
66.	Suspension line attaching loop, greater than 2 inches up to 4 inches in length		
	1/4 inch or less		260
	Greater than 1/4 inch	128a	
67.	Location of reefing line cutter bracket from edge of skirt		
	1/8 inch or less		261
	Greater than 1/8 inch	129a	
68.	Location of reefing line arming cord from edge of skirt		
	1/8 inch or less		262
	Greater than 1/8 inch	130a	
69.	Clear oil stains on canopy cloth (sewing machine oil)		
	Stains greater than 1/4 square inch (or stains greater than 1/2 inch x 1/2 inch, or 1/4 inch x 1 inch, for example)	131a	
	Two to three stains on a single gore section		263
	More than three stains on a single gore section	132a	
	More than a total of 4 square inches per entire canopy	133a	
70.	Material unclean, stain (oil, mold, dirt, etc.)	134a	
71.	Identification marking		
	Incorrect or missing	135a	
	Block stamp or printed text illegible		264
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	136a	
72.	Ends of webbing, tape, or cord not seared, scissor cut, waxed, serged or treated with anti-fray compound as specified		265
73.	Searing over stitching	137a	
74.	Thread ends not trimmed		
	1/4 inch up to 2 inches		266
	Greater than 2 inches	138a	
	Greater than ½ inch at anti-inversion net	139a	

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Table IV. Classification of defects - canopies: ring slot and ribbon parachutes.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Improper assembly of vent reinforcement band	107	
8.	Vent reinforcement band overlap		
	1/4 inch or less, minus		201
	Greater than 1/4 inch, minus	108	
	Greater than1/2 inch, plus		202
9.	Vent loop length (finished dimension)		
	1/2 inch or less		203
	Greater than 1/2 inch	109	
10.	Vent line length (finished dimension)		
	1/4 inch or less		204
	Greater than 1/4 inch	110	
11.	Variation in vent line length per set		
	1 percent or less		205
	Greater than 1 percent	111	
12.	Twisted vent line (flat)	112	
13.	Twisted vent line (core or coreless braid)		
	Greater than 360 degrees		206
	Greater than 360 degrees causing kinks in line when relaxed	113	
14.	Gore width at vent, measured centerline to centerline on adjacent radial seams		
	1/4 inch or less		207
	Greater than 1/4 inch	114	
15.	More or less than the required number of ribbons per gore	115	
16.	Spacing of horizontal ribbons (ribbon canopy)		
	3/16 inch or less		208

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Table IV. Classification of defects - canopies: ring slot and ribbon parachutes - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
	Greater than 3/16 inch	116	
17. Spacing of verticals			
	1/4 inch or less		209
	Greater than 1/4 inch	117	
18. Slot width (ring slot canopy)			
	1/8 inch or less		210
	Greater than 1/8 inch	118	
19. Leading and trailing edge of section hems (ring slot canopies only)			
	Underfolds where gore fabric is not butted flush with fold of hem and caught with two rows of stitches (when specified by drawing)	119	
	Intermittent underfolds up to 2 inches along seam length		211
	Underfold greater than 2 inches along seam length	120	
	Fullness between lines of stitching		
	1/8 inch or less		212
	Greater than 1/8 inch	121	
20. Gore width at skirt, measured centerline to centerline on adjacent radial seams			
	1/4 inch or less		213
	Greater than 1/4 inch	122	
21. Skirt reinforcement band overlap			
	1/4 inch or less, minus		214
	Greater than 1/4 inch, minus	123	
	Greater than 1/2 inch, plus		215
22. Improper assembly of skirt reinforcement band		124	
23. Vent and skirt hems improperly formed			
	Raw edge showing as a result of an underfold	125	
	Intermittent underfold where edge is caught with two rows of stitching of a 4-needle seam		216
	Underfold where edge is caught with one row of stitching	126	
	Raw edge showing as a result of an overfold where the edge extends more than 1 inch from inside edge of hem	127	
24. Vent and shirt hems impropertly formed (two rows of stitching)			
	Raw edge showing	128	
	Intermittent underfold, half of seam width or greater, where edge is caught with one row of stitching	129	
25. Pleats and wrinkles (all hems)			
	1/8 inch up to 1/4 inch and one per gore		217

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Table IV. Classification of defects - canopies: ring slot and ribbon parachutes - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Greater than 1/8 inch and more than one per gore	130	
	Greater than 1/8 inch and more than six per canopy	131	
	Greater than 1/4 inch	132	
26.	Tucks	133	
27.	Vent lines out of sequence	134	
28.	Suspension lines crossed	135	
29.	Cut, frayed, or damaged suspension or vent lines	136	
30.	Radial and vertical seam length		
	1 percent or less		218
	Greater than 1 percent	137	
31.	Variation in radial seam length per canopy		
	1/2 percent or less		219
	Greater than 1/2 percent	138	
32.	Twist in radial seam	139	
33.	Variation in suspension line length per line group		
	1/2 percent or less		220
	Greater than 1/2 percent	140	
34.	Variation in suspension line length per set		
	1 percent or less		221
	Greater than 1 percent	141	
35.	Twisted suspension line (flat), except hem rigged canopy	142	
36.	Twisted suspension line (flat), hem rigged canopy		
	180 up to 360 degrees		222
	Greater than 360 degrees	143	
37.	Vertical tapes not equally spaced		
	1/2 inch up to 1 inch		223
	Greater than 1 inch	144	
38.	Twisted suspension line (core and coreless braid)		
	Greater than 360 degrees		224
	Greater than 360 degrees causing kinks in line	145	
39.	Pocket band free length		

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Table IV. Classification of defects - canopies: ring slot and ribbon parachutes - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	1/8 inch or less		225
	Greater than 1/8 inch	146	
40.	Pocket depth, centerline of radial seam to stitch pattern on skirt band		
	1/8 inch or less		226
	Greater than 1/8 inch	147	
41.	Open seams	148	
42.	Stitching		
	Stitch per inch defects size 3 - 6 thread:		
	One stitch per inch more or less than the number specified, up to 2 inches in length, and one place per seam		227
	More than two stitches per inch more or less than the number specified	149	
	More than one stitch per inch more or less than the number specified, longer than 2 inches in length	150	
	More than one stitch per inch defect per seam	151	
	Stitch per inch defects size E - FF thread:		
	Two stitches per inch more or less than the number specified, up to 2 inches in length, and one place per seam		228
	More than two stitches per inch more or less than the number specified	152	
	More than one stitch per inch more or less than the number specified, longer than 2 inches in length	153	
	More than one stitch per inch defect per seam	154	
	Variation in stitches per inch due to changes in material thickness		
	Specified stitches per inch out of tolerance for 3 inches or less up to two places per seam, or spaced 36 inches apart or greater, or both		229
	Specified stitches per inch out of tolerance for greater than 3 inches more than two places per seam, or spaced closer than 36 inches part, or both	155	
	Stitch pattern defects:		
	Any broken or missing stitch	156	
	Broken, skipped, or missing stitches over 1/2 inch in any row of 4-needle stitching	157	
	One skipped stitch		230
	Two or more consecutive skipped stitches	158	
	Missing or incomplete stitch pattern (for example, bartack or zigzag)	159	
	Backstitch missing	160	
	Backstitch short		
	1/4 inch or less		231
	Greater than 1/4 inch	161	
	Backstitch long, greater than 1/4 inch		232
	Runoff, two or more stitches into single cloth or beyond seam or hem margins	162	
	Zigzag stitching where less than 85 percent of the stitches fully penetrating the suspension line	163	
	Stitch tension too loose or too tight		

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Table IV. Classification of defects - canopies: ring slot and ribbon parachutes - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	3 inches long or less per seam		233
	Greater than 3 inches longper seam	164	
	Kinks, intermittent, spaced closer than 2 inches apart		234
	Stitch pattern dimensions not as specified, stitch patterns 5 inches long or less		
	1/8 inch or less		235
	Greater than 1/8 inch, minus	165	
	Greater than 1/8 inch, plus		236
	Loose or raveled bartack stitching	166	
	Repairs to bartack stitching other than complete removal and replacement	167	
	Birds nest (stitching nest), to include masses of knotted and tangled thread	168	
43.	Frequency of repair to stitching and splicing on canopies, regardless of the number of rows of stitches in the seam in excess of that specified below	169	

Length of seams (inches)	Splices	Repairs
up to 2	0	0
greater than 2 up to 12	0	1
greater than 12 up to 72	0	2
greater than 72 up to 240	1	3
greater than 240 up to 360	2	4
greater than 360 up to 480	3	5
greater than 480 up to 600	4	6
Skirt band	1 per 600 inches	1 per 100 inches

44.	Damage to textiles, not including cuts, tears and holes		
	1/8 inch long or less and one per canopy up to 35 feet in diameter		237
	Greater than 1/8 inch long or more than one per canopy up to 35 feet diameter	170	
	1/8 inch long or less and two per canopy greater than 35 feet diameter		238
	Greater than 1/8 inch long or more than two per canopy greater than 35 feet diameter	171	
45.	Any cut, tear or hole	172	
46.	Darns		
	More than one per gore	173	
	More than ten per canopy up to 50 feet diameter	174	
	More than fifteen per canopy from 50 feet up to 75 feet diameter	175	
	More than twenty per canopy 75 feet diameter and greater	176	
47.	Foldback of suspension lines at connector link or skirt attachment loop		
	1/4 inch or less, minus		239
	Greater than 1/4 inch, minus	177	
	Greater than 1/2 inch, plus		240
48.	Misalignment of cutter pockets	178	

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Table IV. Classification of defects - canopies: ring slot and ribbon parachutes - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
49.	Loop, 2 inches in length or less		
	1/8 inch or less		241
	Greater than 1/8 inch	179	
50.	Loop, 2 inches up to 4 inches in length		
	1/4 inch or less		242
	Greater than 1/4 inch	180	
51.	Dark identification threads not removed or markings not yellow (for special weapon parachutes, other than those identified as training units)	181	
52.	Clear oil stains on canopy cloth (sewing machine oil)		
	Stains greater than 1/4 square inch (or stains greater than 1/2 inch x 1/2 inch, or 1/4 inch x 1 inch, for example)	182	
	Two to three stains on a single gore section		243
	More than three stains on a single gore section	183	
	More than a total of four square inches per entire canopy	184	
53.	Material unclean, stain (oil, mold, dirt, etc.)	185	
54.	Stained or otherwise discolored and/or chaffed sections of material	186	
55.	Identification marking		
	Incorrect or missing	187	
	Block stamp or printed text illegible		244
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	188	
56.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		245
57.	Searing over stitching	189	
58.	Thread ends not trimmed		
	1/4 inch up to 2 inches		246
	Greater than 2 inches	190	

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Table V. Classification of defects - pilot chutes, drogues, and ejector spring assembly.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Impact tape, improper construction or assembly	107	
8.	Radial seam length		
	1/4 inch or less		201
	Greater than 1/4 inch	108	
9.	Gore width at skirt, centerline to centerline on radial seam		
	1/8 inch or less		202
	Greater than 1/8 inch	109	
10.	Skirt webbing overlap		
	1/4 inch or less, minus		203
	Greater than 1/4 inch, minus	110	
	Greater than 1/4 inch, plus		204
11.	Seam folding (raw edge)	111	
12.	Pleats		
	1/16 inch up to 1/8 inch and one per gore		205
	Greater than 1/8 inch or more than one per gore	112	
13.	Variation in suspension line length		
	1/2 percent or less		206
	Greater than 1/2 percent	113	
14.	Suspension line twisted		
	Greater than 360 degrees		207
	Greater than 360 degrees causing kinks in line	114	
15.	Suspension lines fold back or overlap skirt		
	1/8 inch or less, minus		208
	Greater than 1/8 inch, minus	115	
	Greater than 1/4 inch, plus		209

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Table V. Classification of defects - pilot chutes, drogues, and ejector spring assembly - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
16.	Loop length		
	1/8 inch or less, minus		210
	Greater than 1/8 inch, minus	116	
	Greater than 1/4 inch, plus		211
17.	Suspension line out of sequence at the loop collar	117	
18.	Spring tension less than required loading	118	
19.	Spring not hand tacked to pilot chute cone	119	
20.	Open seams	120	
21.	Stitching		
	Two stitches per inch more or less than the number specified, one place per seam		212
	More than two stitches per inch more or less than the number specified	121	
	Two stitches per inch or more, more or less than the number specified, more than one place per seam	122	
	Any broken or missing stitch	123	
	Two or more consecutive skipped stitches	124	
	Missing or incomplete stitch pattern	125	
	Backstitch missing	126	
	Backstitch short		
	Greater than 1/16 inch up to 1/8 inch		213
	Greater than 1/8 inch	127	
	Backstitch long, greater than 1/4 inch		214
	Runoff		
	Three or more stitches into single cloth or beyond seam or hem margins	128	
	1/4 inch or less		215
	Greater than 1/4 inch	129	
	Stitch tension too loose or too tight		
	4 inches or less per seam		216
	Greater than 4 inches per seam	130	
	Kinks, intermittent, spaced closer than 2 inches apart		217
	Stitch pattern dimensions not as specified, stitch patterns 5 inches in length or less		
	1/8 inch or less		218
	Greater than 1/8 inch, minus	131	
	Greater than 1/8 inch, plus		219
	Stitch pattern dimensions not as specified, stitch patterns greater than 5 inches in length		
	1/4 inch or less		220

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Table V. Classification of defects - pilot chutes, drogues, and ejector spring assembly - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Greater than 1/4 inch, minus	132	
	Greater than 1/4 inch, plus		221
	Loose or raveled bartack stitching	133	
	Repairs to bartack stitching other than complete removal and replacement	134	
	Birds nest (stitching nest), to include masses of knotted and tangled thread	135	
22.	Damage to textiles, not including cuts, tears and holes		
	1/8 inch long or less and one per pilot chute		222
	Greater than 1/8 inch long and more than one per pilot chute	136	
23.	Cuts, tears or holes (except necessary drill holes)	137	
24.	Darns		
	More than one darn per canopy	138	
	Any darn on T-11R Extractor	139	
25.	Clear oil stains on canopy cloth (sewing machine oil)		
	Stains greater than 1/4 square inch (or stains greater than 1/2 inch x 1/2 inch, or 1/4 inch x 1 inch, for example)	140	
	More than one stain per pilot chute	141	
26.	Material unclean, stain (oil, mold, dirt, etc.)	142	
27.	Identification marking		
	Incorrect or missing	143	
	Block stamp or printed text illegible		223
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	144	
28.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		224
29.	Searing over stitching	145	
30.	Thread ends not trimmed		
	1/4 inch up to 2 inches		225
	Greater than 2 inches	146	

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TABLE VI. Classification of defects - packs or containers, and deployment bags - personnel.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Seam folding		
	Improperly formed seam or hem exposing raveled edge of fabric	107	
8.	Binding tapes		
	Splicing of binding tape raw edge not turned under 1/4 inch		201
	Improperly attached, exposing raw edge of fabric	108	
9.	Finished dimensions for dimensions 10 inches or greater		
	1/4 inch or less, plus or minus		202
	Greater than 1/4 inch, plus or minus	109	
10.	Grommet, cone and snap fastener spacing and location		
	1/16 inch up to 1/8 inch		203
	Greater than 1/8 inch	110	
11.	Assembly of grommets		
	Improperly set, two or more exposed teeth for spur washers (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	111	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	112	
	Clinched excessively tight, causing cut fabric	113	
	Improper method of installation (for example, base hole in fabric too large)	114	
	Roll of grommet contains splits, burrs or sharp edges, is bent or deformed	115	
12.	Assembly of snaps		
	Location of dot on directional snap fastener misaligned more than 10 degrees	116	
	Loose, broken or sharp edges	117	
	Improper method of installation (for example, base hole in fabric too large)	118	
13.	Stitching		
	Up to two stitches per inch more or less than the number specified, up to 2 inches in length		204

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TABLE VI. Classification of defects - packs or containers, and deployment bags - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
	Up to two stitches per inch more or less than the number specified, longer than 2 inches in length	119	
	More than two stitches per inch more or less than the number specified, longer than 2 inches in length	120	
	Any broken or missing stitch	121	
	Two or more consecutive skipped stitches	122	
	Runoff greater than 1/2 inch in length in any one place or greater than 1-1/2 inch total length per pack	123	
	Runoff on edge binding, greater than 1/2 inch in any one place or greater than 1 inch in length per pack	124	
	Missing or incomplete stitch pattern (for example, bartack or cross box)	125	
	Backstitch short		
	1/4 inch or less		205
	Greater than 1/4 inch	126	
	Backstitch long, greater than 1/4 inch		206
	Tension too loose or too tight, 4 inches or less per seam		207
	Tension too loose or too tight, greater than 4 inches per seam	127	
	Kinks, intermittent, spaced closer than 2 inches apart		208
	Stitch pattern lengths not as specified (stitch patterns 5 inches in length or less)		
	1/8 inch or less		209
	Greater than 1/8 inch, minus	128	
	Greater than 1/8 inch, plus		210
	Stitch pattern lengths not as specified (stitch patterns greater than 5 inches in length)		
	1/4 inch or less		211
	Greater than 1/4 inch, minus	129	
	Greater than 1/4 inch, plus		212
	One point on 3 or 4-point stitching does not carry over edge of webbing		213
	More than one point on 3 or 4-point stitching does not carry over edge of webbing	130	
	Loose or raveled bartack stitching	131	
	Repairs to bartack stitching other than complete removal and replacement	132	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	133	
14. Damage to textiles, not including cuts, tears and holes			
	1/4 inch or less and one per pack		214
	Greater than 1/4 inch, or more than one per pack	134	
15. Cuts, tears, holes (except necessary drill holes)			
	Two per pack less than 3/16 inch diameter, or less than 1/4 inch long		215
	Greater than 3/16 inch in diameter or greater than 1/4 inch long	135	
	More than two per pack	136	
16. Darns			
	Three or less per pack		216

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TABLE VI. Classification of defects - packs or containers, and deployment bags - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	More than three per pack	137	
17.	Overall pleat dimension greater than 1/16 inch		217
18.	Stow loop dimensions		
	23/32 inch and 25/32 inch dimensions		
	1/32 inch up to 1/16 inch, plus, and up to five per bag		218
	1/32 inch up to 1/16 inch, plus, and more than five per bag	138	
	Greater than 1/8 inch, plus	139	
	Greater than 1/16 inch, minus	140	
	2-5/8 inch dimension between stitching		
	1/8 inch or less, plus, and five per bag		219
	1/8 inch or less, plus, and more than five per bag	141	
	Greater than 1/8 inch, plus	142	
	Greater than 1/16 inch, minus	143	
19.	Identification marking		
	Incorrect or missing	144	
	Block stamp or printed text illegible		220
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	145	
20.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		221
21.	Searing over stitching	146	
22.	Thread ends not trimmed		
	1/4 inch up to 2 inches		222
	Greater than 2 inches	147	

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TABLE VII. Classification of defects - harnesses, risers, static lines, bridle lines, straps, lanyards, pockets, and cloth ripcord handles - personnel.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component(s) missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Any webbing splice	107	
8.	Length (finished dimensions)		
	Leg straps, diagonal back straps, and risers		
	1 inch or less		201
	Greater than 1 inch	108	
	Static lines		
	1-1/2 inches or less		202
	Greater than 1-1/2 inch	109	
9.	Symmetry out of tolerance	110	
10.	Assembly of grommets		
	Improperly set, two or more exposed teeth for spur washers (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	111	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	112	
	Clinched excessively tight, causing cut fabric	113	
	Improper method of installation (for example, base hole in fabric too large)	114	
	Roll of grommet contains splits, burrs or sharp edges, is bent or deformed	115	
11.	Twisted back, chest, or leg straps	116	
12.	Stitching		
	One stitch per inch more or less than the number specified, up to 2 inches in length		203
	One stitch per inch more or less than the number specified, longer than 2 inches in length	117	
	Any broken or missing stitch	118	
	More than one stitch reinforcement per stitch formation	119	
	More than three stitch reinforcements per harness	120	
	More than one skipped stitch per stitch pattern	121	

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TABLE VII. Classification of defects - harnesses, risers, static lines, bridle lines, straps, lanyards, pockets, and cloth ripcord handles - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Two or more consecutive skipped stitches	122	
	More than three skipped stitches per harness	123	
	Loops in stitching		
	3/32 inch up to 1/8 inch high		204
	Greater than 1/8 inch high	124	
	Loops greater than one stitch in length	125	
	Loops spaced closer than 2 inches apart	126	
	More than three loops per harness	127	
	Any broken stitch, skipped stitch, missing stitch, loop or stitch repair in 3 or 4-point stitching on static or bridle lines	128	
	Any broken stitch, skipped stitch, missing stitch, loop or stitch repair on risers	129	
	NOTE: One skipped or missing stitch and one stitch repair will be permitted in the riser spreader bar.		
	Missing or incomplete stitch pattern (for example, bartack or cross box)	130	
	Runoff, two or more stitches off webbing	131	
	Missing backstitch	132	
	Backstitch short		
	Greater than 1/16 inch up to 1/8 inch		205
	Greater than 1/8 inch	133	
	Backstitch long, greater than 1/4 inch		206
	Tension too loose or too tight in any one row		
	2 inches or less		207
	Greater than 2 inches	134	
	Kinks, intermittent, spaced closer than 2 inches apart		208
	Stitch pattern length not as specified (stitch patterns 5 inches in length or less)		
	1/8 inch or less		209
	Greater than 1/8 inch, minus	135	
	Greater than 1/8 inch, plus		210
	Stitch pattern lengths not as specified (stitch patterns greater than 5 inches in length)		
	1/4 inch or less		211
	Greater than 1/4 inch, minus	136	
	Greater than 1/4 inch, plus		212
	One point on 3 or 4-point stitching does not carry over edge of webbing		213
	More than one point on 3 or 4-point stitching does not carry over edge of webbing	137	
	Loose or raveled bartack stitching	138	
	Repairs to bartack stitching other than complete removal and replacement	139	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	140	
13.	Damage to textiles, not including cuts, tears and holes		
	One cut warp end		214
	More than one cut warp end	141	
	Damage to warp or filling yarn greater than 3/8 inch long or 1/8 inch wide	142	
	Damage to more than one yarn on the risers or static and bridle line webbing	143	

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TABLE VII. Classification of defects - harnesses, risers, static lines, bridle lines, straps, lanyards, pockets, and cloth ripcord handles - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Damage to more than three filling yarns on the harness	144	
14.	Any darn or patch NOTE: This defect does not apply to vest panels on Restraint Harness assemblies	145	
15.	Repairs		
	Broken or missing stitching not repaired by reinforcement	146	
	Repairs to risers	147	
	Repairs to 3 or 4-point stitching on static or bridle lines	148	
16.	Identification marking		
	Incorrect or missing	149	
	Block stamp or printed text illegible		215
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	150	
17.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		216
18.	Searing over stitching	151	
19.	Thread ends not trimmed		
	1/4 inch up to 2 inches		217
	Greater than 2 inches	152	

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TABLE VIII. Classification of defects - deployment bags and packs - other than personnel.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Seam folding improperly formed or hem exposing raw edge of fabric	107	
8.	Binding tapes		
	Splicing of binding tape raw edge not turned under 1/4 inch		201
	Improperly attached, exposing raw edge of fabric	108	
9.	Overall length, width, depth and circumference out of tolerance		
	1/4 inch or less		202
	Greater than 1/4 inch	109	
10.	Grommet or snap fastener spacing		
	1/16 inch up to 1/8 inch		203
	Greater than 1/8 inch	110	
11.	Assembly of grommets		
	Improperly set, two or more exposed teeth for spur washers (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	111	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	112	
	Clinched excessively tight, causing cut fabric	113	
	Improper method of installation (for example, base hole in fabric too large)	114	
	Roll of grommet contains splits, burrs or sharp edges, is bent or deformed	115	
12.	Stitching		
	Two stitches per inch more or less than the number specified, up to 2 inches in length		204
	Two stitches per inch more or less than the number specified, and longer than 2 inches in length	116	
	Any broken or missing stitch	117	
	Two or more consecutive skipped stitches	118	
	Missing or incomplete stitch pattern (for example, bartack or cross box)	119	
	Missing backstitch	120	

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TABLE VIII. Classification of defects - deployment bags and packs - other than personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Backstitch short		
	1/4 inch or less		205
	Greater than 1/4 inch	121	
	Backstitch long, greater than 1/4 inch		206
	Runoff on edge binding, greater than 1/2 inch in length or greater than 1 inch in length per pack	122	
	Tension too loose or too tight, 4 inches or less per seam		207
	Tension too loose or too tight, greater than 4 inches per seam	123	
	Kinks, intermittent, spaced closer than 2 inches apart		208
	Stitch pattern lengths not as specified (stitch patterns 5 inches in length or less)		
	1/8 inch or less		209
	Greater than 1/8 inch, minus	124	
	Greater than 1/8 inch, plus		210
	Stitch pattern lengths not as specified (stitch patterns greater than 5 inches in length)		
	1/4 inch or less		211
	Greater than 1/4 inch, minus	125	
	Greater than 1/4 inch, plus		212
	Defective stitching not repaired correctly or exceeding specified limits	126	
	Loose or raveled bartack stitching	127	
	Repairs to bartack stitching other than complete removal and replacement	128	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	129	
	Missing or incomplete stitch pattern	130	
13.	Unintended pleats or folds greater than 1/16 inch	131	
14.	Damage to textiles, not including cuts, tears and holes		
	1/4 inch or less and two per bag		213
	Greater than 1/4 inch or more than two per bag	132	
15.	Cuts, tears, or holes (except necessary drill holes)		
	1/16 inch diameter or less, up to three per bag		214
	Greater than 1/16 inch diameter	133	
	More than three per bag, any size	134	
16.	Darns and patches		
	Darns		
	1/4 inch up to 1/2 inch		215
	Greater than 1/2 inch	135	
	Darns, more than four per bag, any size	136	
	More than one patch per bag	137	
	Patches exceeding an area of 36 square inches	138	
	Patches where one side exceeds 8 inches in length	139	
17.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with		216

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TABLE VIII. Classification of defects - deployment bags and packs - other than personnel - Continued.

No.	Classification of defects	Classification	
		Major/Critical	Minor
	anti-fray compound as specified		
18.	Identification marking		
	Incorrect or missing	140	
	Block stamp or printed text illegible		217
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	141	
19.	Searing over stitching	142	
20.	Thread ends not trimmed		
	1/4 inch up to 2 inches		218
	Greater than 2 inches	143	

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TABLE IX. Classification of defects - risers, static lines, bridles, straps, adapter webs, and retainers - other than personnel.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Twisted or malformed part	107	
8.	Assembly of grommets		
	Improperly set, two or more exposed teeth for spur washers (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	108	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	109	
	Clinched excessively tight, causing cut fabric	110	
	Improper method of installation (for example, base hole in fabric too large)	111	
	Roll of grommet contains splits, burrs or sharp edges, is bent or deformed	112	
9.	Overall length of item out of tolerance		
	1 percent or less		201
	Greater than 1 percent	113	
10.	Spacing dimensions		
	1 percent or less		202
	Greater than 1 percent	114	
11.	Designed textile loop, 2 inches in length or less		
	1/8 inch or less		203
	Greater than 1/8 inch	115	
12.	Designed textile loop, greater than 2 inches up to 4 inches in length		
	1/4 inch or less		204
	Greater than 1/4 inch	116	
13.	Stitching		
	Two stitches per inch more or less than the number specified, up to 4 inches in length		205
	Two stitches per inch more or less than the number specified, longer than 2 inches in length	117	

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TABLE IX. Classification of defects - risers, static lines, bridles, straps, adapter webs, and retainers - other than personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Broken stitches	118	
	Skipped stitches		
	1/4 inch or less in length		206
	Greater than 1/4 inch in length	119	
	Loops in stitching		
	1/8 inch up to 1/4 inch		207
	Greater than 1/4 inch	120	
	Missing or incomplete stitch pattern	121	
	Backstitch short		
	1/4 inch or less		208
	Greater than 1/4 inch	122	
	Backstitch long, greater than 1/4 inch		209
	Tension too loose or too tight, 4 inches or less per seam		210
	Tension too loose or too tight, greater than 4 inches per seam	123	
	Stitch pattern length not as specified (stitch patterns 5 inches in length or less)		
	1/8 inch or less		211
	Greater than 1/8 inch, minus	124	
	Greater than 1/8 inch, plus		212
	Stitch pattern length not as specified (stitch patterns greater than 5 inches in length)		
	1/4 inch or less		213
	Greater than 1/4 inch, minus	125	
	Greater than 1/4 inch, plus		214
	Defective stitching not repaired correctly or exceeding specified limits	126	
	Loose or raveled bartack stitching	127	
	Repairs to bartack stitching other than complete removal and replacement	128	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	129	
14.	Dark identification threads not removed or markings not yellow (for special weapon parachute systems other than those identified as training units)	130	
15.	Material unclean, stain (oil, mold, dirt, etc.)	131	
16.	Specified Identification markings		
	Incorrect or missing	132	
	Block stamp or printed text illegible		215
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	133	
17.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		216
18.	Searing over stitching	134	
19.	Thread ends not trimmed		

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TABLE IX. Classification of defects - risers, static lines, bridles, straps, adapter webs, and retainers - other than personnel - Continued.

No.	Classification of defects	Classification	
		Major/Critical	Minor
	1/4 inch up to 2 inches		217
	Greater than 2 inches	135	

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TABLE X. Classification of defects - integrated pack and harness - personnel.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Seam folding		
	Improperly formed seam or hem exposing raw edge of fabric	107	
8.	Binding tapes		
	Splicing of binding tape raw edge not turned under 1/4 inch		201
	End of binding tape improperly finished		202
	Corners of binding tape improperly formed, fold not stitched down		203
9.	Grommet, cone and snap fastener spacing and location		
	1/16 inch up to 1/8 inch		204
	Greater than 1/8 inch	108	
10.	Assembly of grommets		
	Improperly set, two or more exposed teeth for spur washers (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	109	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	110	
	Clinched excessively tight, causing cut fabric	111	
	Improper method of installation (for example, base hole in fabric too large)	112	
	Roll of grommet contains splits, burrs or sharp edges, is bent or deformed	113	
11.	Assembly of snaps		
	Location of dot on directional snap fastener misaligned more than 10 degrees	114	
	Loose, broken or sharp edges	115	
	Improper method of installation (for example, base hole in fabric too large)	116	
12.	Assembly of cones		
	Improper method of installation, not in accordance with TDP	117	
	Broken or sharp edges	118	
13.	Slide fasteners installed improperly or incorrectly	119	

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TABLE X. Classification of defects - integrated pack and harness - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
14.	Loop, 2 inches in length or less		
	1/8 inch or less		205
	Greater than 1/8 inch	120	
15.	Loop, greater than 2 inches in length		
	3/16 inch or less		206
	Greater than 3/16 inch	121	
16.	Location of deployment gun attachment point		
	1/16 inch or less		207
	Greater than 1/16 inch	122	
17.	Location of clevis assembly on flap		
	1/32 inch or less		208
	Greater than 1/32 inch	123	
18.	Location of rip cord housing clamp		
	1/16 inch or less		209
	Greater than 1/16 inch	124	
19.	Any hole (except necessary drill holes)	125	
20.	Darns (bag, flap or pad)		
	More than four darns, any size	126	
	One darn on hole up to 1/4 inch diameter or tear up to 1/2 inch in length		210
	Any darn or repair on a hole greater than 1/4 inch diameter or tear greater than 1/2 inch in length	127	
21.	Any other darn	128	
22.	Any patch	129	
23.	Variation in length of leg straps		
	1 inch or less		211
	Greater than 1 inch	130	
24.	Item of hardware missing	131	
25.	Twisted chest or leg straps	132	
26.	Lanyard length		
	1/4 inch or less		212
	Greater than 1/4 inch	133	
27.	Stitching		

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TABLE X. Classification of defects - integrated pack and harness - personnel - Continued.

No.	Classification of defects	Classification	
		Major/Critical	Minor
	One stitch per inch more or less than the number specified, 2 inches in length or less		213
	One stitch per inch more or less than the number specified, greater than 2 inches in length	134	
	Any broken or missing stitch	135	
	Two or more consecutive skipped stitches	136	
	One skipped stitch on risers	137	
	NOTE: One skipped stitch on the back strap or leg strap will be permitted.		
	Four or more skipped stitches per harness	138	
	Loops in stitching due to stitching over varying thicknesses of webbing		
	3/8 inch or less		214
	Greater than 3/8 inch	139	
	Missing or incomplete stitch pattern (for example, bartack or cross box)	140	
	Runoff, two or more stitches off webbing	141	
	Missing backstitch	142	
	Backstitch short		
	Greater than 1/16 inch up to 1/8 inch		215
	Greater than 1/8 inch	143	
	Backstitch long, greater than 1/4 inch		216
	Tension too loose or too tight in any one row		
	2 inches or less		217
	Greater than 2 inches	144	
	Kinks, intermittent, spaced closer than 2 inches apart		218
	Stitch pattern length not as specified (stitch patterns 5 inches in length or less)		
	1/8 inch or less		219
	Greater than 1/8 inch, minus	145	
	Greater than 1/8 inch, plus		220
	Stitch pattern length not as specified (stitch patterns greater than 5 inches in length)		
	1/4 inch or less		221
	Greater than 1/4 inch, minus	146	
	Greater than 1/4 inch, plus		222
	Defective stitching not repaired correctly or exceeding specified limits	147	
	Loose or raveled bartack stitching	148	
	Repairs to bartack stitching other than complete removal and replacement	149	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	150	
28.	Damage to textiles, not including cuts, tears and holes		
	One cut warp end		223
	More than one cut warp end	151	
	Damage to warp or filling yarn greater than 3/8 inch	152	
	Damage to more than one yarn in risers or leg and chest straps	153	
29.	Broken or missing stitching not reinforced by restitiching	154	
	NOTE: Repairs in risers and leg straps not permitted.		

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TABLE X. Classification of defects - integrated pack and harness - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
30.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		
	Used as reinforcement and binding		224
	Used in harness webbing, chest and leg straps	155	
31.	Identification marking		
	Incorrect or missing	156	
	Block stamp or printed text illegible		225
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	157	
32.	Searing over stitching	158	
33.	Thread ends not trimmed		
	1/4 inch up to 2 inches		226
	Greater than 2 inches	159	

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TABLE XI. Classification of defects - canopies: square, gliding type, main and reserve - personnel.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Improper cascade continuity and suspension line routing continuity, one line or greater	107	
8.	Grommet, cone and snap fastener spacing and location		
	1/16 inch up to 1/8 inch		201
	Greater than 1/8 inch	108	
9.	Assembly of grommets		
	Improperly set, two or more exposed teeth for spur washers (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	109	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	110	
	Clinched excessively tight, causing cut fabric	111	
	Improper method of installation (for example, base hole in fabric too large)	112	
	Roll of grommet contains splits, burrs or sharp edges, is bent or deformed	113	
10.	Magnets improperly assembled or installed, and/or do not function properly	114	
11.	Seams not constructed as specified		
	Any seam	115	
	Raw edge exposed, unless otherwise depicted in the TDP	116	
	Open seam	117	
12.	Cut, frayed or damaged suspension lines	118	
13.	Twisted suspension line between canopy and cascade		
	Greater than 180 degrees up to 360 degrees		202
	Greater than 360 degrees	119	
14.	Twisted suspension line between cascade and connector link		
	Greater than 180 degrees up to 360 degrees		203
	Greater than 360 degrees	120	

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TABLE XI. Classification of defects - canopies: square, gliding type, main and reserve - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
15.	Twisted suspension line between canopy and connector link		
	Greater than 180 degrees up to 360 degrees		204
	Greater than 360 degrees	121	
16.	Twisted control line between canopy trailing edge and upper cascade		
	Greater than 180 degrees up to 360 degrees		205
	Greater than 360 degrees	122	
17.	Twisted control line between upper and lower cascade greater than 90 degrees	123	
18.	Reinforcement tape on ribs		
	Take-up greater than 1/4 inch	124	
19.	Rib leading edge tape, RA-1 only		
	Any one rib greater than 1/4 inch, plus or minus	125	
	More than four ribs per canopy out of tolerance	126	
	NOTE: Up to four ribs 1/4 inch or less, plus or minus, will not be counted as a defect. Symmetry requirement must be maintained.		
20.	Reinforcement tape and suspension line tape, rib panel connection		
	Missing	127	
	Improperly attached	128	
	NOTE: The stitch used to secure the reinforcement tape within the seam shall not cross the primary row of double needle seam stitching. The primary row of stitching shall be understood as the immediate stressed row when load is placed upon the seam.		
21.	Heat sear of cross port vent holes improperly formed		
	One per rib		206
	More than six per canopy	129	
22.	Splicing		
	On canopy fabric	130	
	On suspension or control lines	131	
	Webbing or tape		207
	Webbing or tape any location other than trailing edge tape, RA-1 only	132	
	Trailing edge tape; more than two splices or any splice not positioned or formed as allowed in the TDP, RA-1 only	133	
23.	Stitching		
	301 and 308 stitching not in accordance with ASTM D6193, more than one stitch per inch more or less than the number specified		
	2 inches or less		208
	Greater than 2 inches	134	
	Bartack stitching not as specified	135	

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TABLE XI. Classification of defects - canopies: square, gliding type, main and reserve - personnel - Continued.

No.	Classification of defects	Classification	
		Major/Critical	Minor
	Bartack stitch pattern length greater than 1/16, plus, RA-1 suspension and steering lines only	136	
	Any broken or missing stitch	137	
	One skipped stitch		209
	Two or more consecutive skipped stitches	138	
	Missing or incomplete stitch pattern (for example, bartacks or cross box)	139	
	Backstitch missing	140	
	Incorrect backstitch, if specified	141	
	Backstitch short		
	Greater than 1/16 up to 1/8 inch		210
	Greater than 1/8 inch	142	
	Backstitch long, greater than 1/4 inch		211
	Runoff, two or more stitches into single cloth or beyond seam or hem margins	143	
	Tension too loose or too tight		
	3 inches or less per seam		212
	Greater than 3 inches per seam	144	
	Kinks, intermittent, spaced closer than 2 inches apart		213
	Stitch pattern length not as specified	145	
	Defective stitching repaired incorrectly, or exceeding specified limits	146	
	Frequency of repairs to stitching exceeding specified limits	147	
	Resewn stitch patterns and bartacks exceeding specified limits	148	
	Loose or raveled bartack stitching	149	
	Repairs to bartack stitching other than complete removal and replacement	150	
	Bird nest (stitching nest) including masses of knotted and tangled thread	151	
	Any damage to material resulting from stitch rework or repair	152	
24.	Damage to textiles, not including cuts, tears and holes		
	1/8 inch long or less and one per canopy		214
	Greater than 1/8 inch long or more than one per canopy	153	
25.	Cuts, tears and holes (except necessary drill holes)	154	
26.	Darns		
	More than specified	155	
	On slider	156	
	Any darn on RA-1	157	
27.	Any hot tack	158	
28.	Material unclean, stain (oil, mold, dirt, etc.)	159	
29.	Clear oil stains on canopy cloth (sewing machine oil)		
	Stains greater than 1/4 square inch (or stains greater than 1/2 inch x 1/2 inch, or 1/4 inch x 1 inch, for example)	160	

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TABLE XI. Classification of defects - canopies: square, gliding type, main and reserve - personnel - Continued.

No.	Classification of defects	Classification	
		Major/Critical	Minor
	Two to three stains on a single panel		215
	More than three stains on a single panel	161	
	More than a total of 4 square inches per entire canopy	162	
30.	Canopy out of trim	163	
31.	Symmetry out of tolerance between lengths of common line groups greater than specified in TDP	164	
32.	Ends specified as scissor cut but are heat seared	165	
33.	Cloth not heat seared if specified	166	
34.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		216
35.	Identification marking		
	Incorrect or missing	167	
	Block stamp or printed text illegible		217
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	168	
36.	Searing over stitching	169	
37.	Thread ends not trimmed		
	1/4 inch up to 2 inches		218
	Greater than 2 inches	170	

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TABLE XII. Classification of defects – pilot chutes, drogues, and ejector spring assembly - MC-4 and RA-1 only.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Assembly of grommets		
	Improperly set, two or more exposed teeth for spur washers (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	107	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	108	
	Clinched excessively tight, causing cut fabric	109	
	Improper method of installation (for example, base hole in fabric too large)	110	
	Roll of grommet contains splits, burrs or sharp edges, is bent or deformed	111	
8.	Seams not constructed as specified		
	Any seam	112	
	Raw edge exposed, unless otherwise depicted in the TDP	113	
	Open seam	114	
9.	Any hot tack	115	
10.	Overall pleat dimension greater than 3/16 inch wide	116	
11.	Stitching		
	301 and 308 stitching in accordance with ASTM D6193, more than one stitch per inch more or less than the number specified		
	2 inches or less		201
	Greater than 2 inches	117	
	Any broken or missing stitch	118	
	One skipped stitch		202
	Two or more consecutive skipped stitches	119	
	Missing or incomplete stitch pattern (for example, bartack or cross box)	120	
	Stitch pattern lengths not as specified	121	
	Bartack stitching not as specified in applicable drawing	122	
	Bartack stitch pattern lengths not as specified	123	
	Backstitch missing	124	
	Incorrect backstitch, if specified		

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TABLE XII. Classification of defects – pilot chutes, drogues, and ejector spring assembly - MC-4 and RA-1 only - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Single Leg Backstitch if Backstitch or Double Backstitch is specified	125	
	Backstitch if Double Backstitch is specified	126	
	NOTE: If Backstitch is used in lieu of Single Leg Backstitch or Double Backstitch is used in lieu of Backstitch, no defect.		
	NOTE: Backstitch, Single Leg Backstitch and Double Backstitch are specified in the RA-1 TDP. Figures for each can be found in the Manufacturing Specification for RA-1 Parachute System.		
	Backstitch short		
	Greater than 1/16 inch up to 1/8 inch		203
	Greater than 1/8 inch	127	
	Backstitch long, greater than 1/4 inch		204
	Runoff, two or more stitches into single cloth or beyond seam or hem margins	128	
	Tension too loose or too tight		
	3 inches or less per seam		205
	Greater than 3 inches per seam	129	
	Kinks, intermittent, spaced closer than 2 inches apart		206
	Defective stitching not repaired correctly or exceeding specified limits	130	
	Frequency of repairs to stitching exceeding specified limits	131	
	Resewn stitch patterns and bartacks exceeding specified limits	132	
	Loose or raveled bartack stitching	133	
	Repairs to bartack stitching other than complete removal and replacement	134	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	135	
	Any damage to material resulting from stitch rework or repair	136	
12.	Grommet, cone and snap fastener spacing and location		
	1/16 inch up to 1/8 inch		207
	Greater than 1/8 inch	137	
13.	Damage to textiles, not including cuts, tears and holes		
	1/8 inch long or less and one per pilot chute		208
	Greater than 1/8 inch long and more than one per pilot chute	138	
14.	Cuts, tears and holes (except necessary drill holes)	139	
15.	Any darn	140	
16.	Clear oil stains on canopy cloth (sewing machine oil)		
	Stains greater than 1/4 square inch (or stains greater than 1/2 inch x 1/2 inch, or 1/4 inch x 1 inch, for example)	141	
	More than one stain per pilot chute	142	
17.	Material unclean, stain (oil, mold, dirt, etc.)	143	
18.	Spring tension less than required loading per TDP	144	

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TABLE XII. Classification of defects – pilot chutes, drogues, and ejector spring assembly - MC-4 and RA-1 only - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
19.	Spring not hand tacked to pilot chute protection cap	145	
20.	Identification marking		
	Incorrect or missing	146	
	Block stamp or printed text illegible		209
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	147	
21.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		210
22.	Searing over stitching	148	
22.	Thread ends not trimmed		
	1/4 inch up to 2 inches		211
	Greater than 2 inches	149	

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TABLE XIII. Classification of defects – integrated pack and container (without harness), deployment bag and pocket – MC-4 and RA-1 only.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Assembly of grommets		
	Improperly set, two or more exposed teeth for spur washers (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	107	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	108	
	Clinched excessively tight, causing cut fabric	109	
	Improper method of installation (for example, base hole in fabric too large)	110	
	Roll of grommet contains splits, burrs or sharp edges, is bent or defromed	111	
8.	Magnets improperly assembled or installed, and/or do not function properly	112	
9.	Seams not constructed as specified in applicable drawing		
	Any seam	113	
	Raw edge exposed, unless otherwise depicted in the TDP	114	
	Open seam	115	
10.	Overall pleat dimension exceeding specified limits		201
11.	Pleats greater than 1/16 inch in width, MC-4 only		202
12.	Stitching		
	301 and 308 stitching not in accordance with ASTM D6193, more than one stitch per inch more or less than the number specified		
	2 inches or less		203
	Greater than 2 inches	116	
	Bartack stitching not as specified in applicable drawing	117	
	Bartack stitch pattern lengths not as specified, RA-1 only		
	Length greater than 1/16 inch, plus	118	
	Width greater than 1/32 inch, plus	119	
	Any broken or missing stitch	120	
	Two or more consecutive skipped stitches	121	
	Missing or incomplete stitch pattern (for example, bartack or cross box)	122	

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TABLE XIII. Classification of defects – integrated pack and container (without harness), deployment bag and pocket – MC-4 and RA-1 only - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Backstitch missing	123	
	Incorrect backstitch, if specified		
	Single Leg Backstitch if Backstitch or Double Backstitch is specified	124	
	Backstitch if Double Backstitch is specified	125	
	NOTE: If Backstitch is used in lieu of Single Leg Backstitch, or Double Backstitch is used in lieu of Backstitch, no defect.		
	NOTE: Backstitch, Single Leg Backstitch and Double Backstitch are specified in the RA-1 TDP. Figures for each can be found in the Manufacturing Specification for RA-1 Parachute System.		
	Backstitch short		
	Greater than 1/16 inch up to 1/8 inch		204
	Greater than 1/8 inch	126	
	Backstitch long, greater than 1/4 inch		205
	Runoff, two or more stitches into single cloth or beyond seam or hem margins	127	
	Tension too loose or too tight		
	3 inches or less per seam		206
	Greater than 3 inches per seam	128	
	Kinks, intermittent, spaced closer than 2 inches apart		207
	Stitch pattern lengths not in accordance with ASTM D6193 or applicable drawing	129	
	Defective stitching not repaired correctly or exceeding specified limits	130	
	Frequency of repairs to stitching exceeding specified limits	131	
	Resewn stitch patterns and bartacks exceeding specified limits	132	
	Loose or raveled bartack stitching	133	
	Repairs to bartack stitching other than complete removal and replacement	134	
	Any damage to material resulting from stitch rework or repair	135	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	136	
13. Grommet, cone and snap fastener spacing and location			
	1/16 inch up to 1/8 inch		208
	Greater than 1/8 inch	137	
14. Snap fasteners			
	Loose, broken or sharp edges	138	
	Improper method of installation (for example base hole in fabric too large)	139	
15. Damage to textiles, not including cuts, tears and holes			
	1/4 inch long or less and one per pack and container		209
	Greater than 1/4 inch, or more than one per pack and container	140	
16. Cuts, tears or holes (except necessary drill holes)		141	
17. Any darn		142	

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TABLE XIII. Classification of defects – integrated pack and container (without harness), deployment bag and pocket – MC-4 and RA-1 only - Continued.

No.	Classification of defects	Classification	
		Major/Critical	Minor
18.	Cloth not heat seared as specified	143	
19.	Material unclean, stain (oil, mold, dirt, etc.)	144	
20.	Identification marking		
	Incorrect or missing	145	
	Block stamp or printed text illegible		210
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	146	
21.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		211
22.	Searing over stitching	147	
23.	Thread ends not trimmed		
	1/4 inch up to 2 inches		212
	Greater than 2 inches	148	

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TABLE XIV. Classification of defects – harness, riser, bridle line, strap, and lanyard – MC-4 and RA-1 only.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Assembly of grommets		
	Improperly set, two or more exposed teeth for spur washers (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	107	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	108	
	Clinched excessively tight, causing cut fabric	109	
	Improper method of installation (for example, base hole in fabric too large)	110	
	Roll of grommet contains splits, burrs or sharp edges, is bent or deformed	111	
8.	Snap fasteners		
	Loose, broken or sharp edges	112	
	Improper method of installation (for example base hole in fabric too large)	113	
9.	Stitching		
	Stitching with size 5 thread:		
	301 stitching not in accordance with ASTM D6193, stitches per inch not as specified		
	1 inch or less		201
	Greater than 1 inch	114	
	Stitch pattern dimensions not in accordance with applicable drawing		
	Length greater than 1/8 inch, plus	115	
	Width greater than 1/16 inch, plus	116	
	Missing or incomplete stitch pattern (for example 3 or 4-point)	117	
	Broken or missing stitches	118	
	One point on 3 or 4-point stitching does not carry over edge of webbing		202
	Two or more consecutive skipped stitches	119	
	More than one skipped stitch per stich pattern on harness	120	
	More than three skipped stitches per harness	121	
	Any skipped stitch on main or reserve risers	122	
	Tension too loose or too tight	123	
	Loops greater than 3/32 inch high or longer than one stitch in length	124	
	Loops, spaced closer than 2 inches apart	125	

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TABLE XIV. Classification of defects – harness, riser, bridle line, strap, and lanyard – MC-4 and RA-1 only - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Loops on back and leg straps	126	
	More than three loops per harness	127	
	Loops in stitching due to stretching over varying thicknesses of webbing		
	3/16 inch up to 3/8 inch		203
	Greater than 3/8 inch	128	
	Kinks		
	Spaced closer than 1 inch apart	129	
	More than specified	130	
	Defective stitching not repaired correctly	131	
	Stitch repairs exceeding specified limits	132	
	Bird nest (stitching nest) to include masses of knotted and tangles thread	133	
	Stitching with size E and size FF thread:		
	301 and 308 stitching not in accordance with ASTM D6193, more than one stitch per inch greater than or less than the number specified		
	2 inches or less		204
	Greater than 2 inches	134	
	Backstitch missing	135	
	Incorrect backstitch		
	Single Leg Backstitch if Backstitch or Double Backstitch is specified	136	
	Backstitch if Double Backstitch is specified	137	
	NOTE: If Backstitch is used in lieu of Single Leg Backstitch or Double Backstitch is used in lieu of Backstitch, no defect.		
	NOTE: Backstitch, Single Leg Backstitch and Double Backstitch are specified in the RA-1 TDP. Figures for each can be found in the Manufacturing Specification for RA-1 Parachute System.		
	Backstitch short		
	Greater than 1/16 inch up to 1/8 inch		205
	Greater than 1/8 inch	138	
	Backstitch long, greater than 1/4 inch		206
	Stitch pattern length not as specified	139	
	Missing or incomplete stitch pattern (for example, bartack or cross box)	140	
	Bartack stitch pattern length not as specified		
	Length, greater than 1/16 inch, plus	141	
	Width, greater than 1/32 inch, plus	142	
	Repairs to bartack stitching other than complete removal and replacement	143	
	Loose or raveled bartack stitching	144	
	Broken or missing stitches	145	
	Two or more consecutive skipped stitches	146	
	Runoff, two or more stitches into single cloth or beyond seam or hem margins	147	
	Tension too loose or too tight		
	3 inches or less per seam		207
	Greater than 3 inches per seam	148	
	Kinks, intermittent, spaced closer than 2 inches apart		208
	Defective stitching not repaired correctly or exceeding specified limits	149	

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TABLE XIV. Classification of defects – harness, riser, bridle line, strap, and lanyard – MC-4 and RA-1 only - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Any damage to material resulting from stitch rework or repair	150	
	Frequency of repairs to stitching exceeding specified limits	151	
	Resewn stitch patterns and bartacks exceeding specified limits	152	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	153	
10.	Damage to textiles, not including cuts, tears and holes		
	One cut warp end		209
	More than one cut warp end	154	
	Damage to filling yarn over 3/8 inch	155	
	NOTE: Damage to more than one yarn not allowed on risers or leg and chest strap.		
11.	Cuts, tears or holes (except necessary drill holes)	156	
12.	Any darn	157	
13.	Material unclean, stain (oil, mold, dirt, etc.)	158	
14.	Symmetry of riser pairs not in accordance with TDP	159	
15.	Twisted back, chest, or leg strap	160	
16.	Hand stitching of tacking not as specified	161	
17.	Identification marking		
	Incorrect or missing	162	
	Block stamp or printed text illegible		210
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	163	
18.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		211
19.	Searing over stitching	164	
20.	Thread ends not trimmed		
	1/4 inch up to 2 inches		212
	Greater than 2 inches	165	

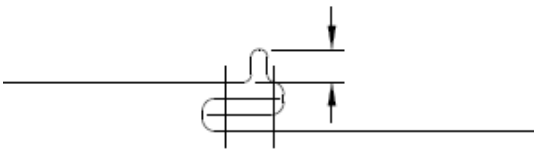
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TABLE XV. Classification of defects – canopies: solid and extended skirt parachute – cargo.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Improper assembly of vent reinforcement band	107	
8.	Vent reinforcement band overlap		
	1/2 inch or less, minus		201
	Greater than 1/2 inch, minus	108	
	Greater than 1 inch, plus		202
9.	Vent hem outside stitching margin from edge of webbing		
	1/16 inch up to 1/4 inch		203
	Greater than 1/4 inch	109	
	All rows of stitching not through reinforcement webbing	110	
10.	Vent loop length (finished dimension)		
	3/4 inch or less, minus		204
	Greater than 3/4 inch, minus	111	
	Greater than 1/2 inch, plus	112	
11.	Vent line length (finished dimension)		
	3/8 inch or less		205
	Greater than 3/8 inch	113	
12.	Variation in vent line length per set		
	1 percent or less		206
	Greater than 1 percent	114	
13.	Twisted vent line (flat)		
	180 degrees or less		207
	Greater than 180 degrees	115	
14.	Twisted vent line (core or coreless braid)		
	Greater than 360 degrees, causing kinks in line when relaxed	116	
15.	Gore width at vent, centerline to centerline on main seam		

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TABLE XV. Classification of defects – canopies: solid and extended skirt parachute – cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	1/4 inch or less		208
	Greater than 1/4 inch	117	
16.	Gore width at skirt, centerline to centerline on main seam		
	1/2 inch or less		209
	Greater than 1/2 inch	118	
17.	Improper type or improper assembly of cross and main seams	119	
18.	Open seam	120	
19.	Underfold (raw edge in main or cross seams)	121	
20.	Overfold in main seam		
	1/2 inch or less or extended into center channel		210
	Greater than 1/2 inch	122	
21.	Underfold in main seam		
	A 4-needle seam not caught by three rows of stitches by at least 1/16 inch	123	
	Three intermittent underfolds of a 2-needle seam, 2 inches long or less and spaced closer than 12 inches apart		211
	More than three intermittent underfolds 1/4 inch wide or greater than 2 inches long, spaced closer than 12 inches apart	124	
	Underfold greater than 6 inches in length	125	
22.	Overfold and underfold in cross seam		
	Underfold 1/8 inch up to 3/16 inch regardless of length		212
	Overfold greater than 1/4 inch regardless of length		213
23.	Fullness in seam width, as measured per the following diagram: 		
	Main seam		
	1/8 inch up to 1/2 inch		214
	Greater than 1/2 inch	126	
	Cross seam		
	1/8 inch up to 1/4 inch		215
	Greater than 1/4 inch	127	
24.	Wrinkles and pleats in cross and main seams		
	Cross seams		
	1/16 inch up to 1/4 inch and one per seam		216
	1/16 inch up to 1/4 inch and more than one per seam	128	

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TABLE XV. Classification of defects – canopies: solid and extended skirt parachute – cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	1/16 inch up to 1/4 inch and more than six per canopy	129	
	Greater than 1/2 inch	130	
	Main seams (canopies up to 100 feet in diameter)		
	1/16 inch up to 1/4 inch and one per seam		217
	1/16 inch up to 1/4 inch and more than one per seam	131	
	1/16 inch up to 1/4 inch and more than six per canopy	132	
	Greater than 1/2 inch	133	
25.	Vent and skirt hems improperly formed		
	Raw edge	134	
	Underfold where edge is caught with one row of stitching		218
26.	Matching of centerlines of seams when vent and skirt hems are folded		
	1/8 inch up to 1/4 inch		219
	Greater than 1/4 inch	135	
27.	Wrinkles, pleats, or folds in skirt and vent hems that do not extend into the gore fabric		
	1/16 inch up to 1/4 inch and up to two per gore		220
	Greater than 1/4 inch and more than two per gore	136	
	Greater than 1/4 inch and more than twelve per canopy, 65 feet in diameter or less	137	
	Greater than 1/4 inch and more than eighteen per canopy, greater than 65 feet in diameter	138	
28.	Wrinkles, pleats or folds in skirt and vent hems extending into gore fabric		
	1/16 inch up to 1/4 inch and two per gore or less		221
	Greater than 1/4 inch and more than two per gore	139	
	Greater than 1/4 inch and more than twelve per canopy, greater than 65 feet in diameter	140	
	Greater than 1/4 inch and more than eighteen per canopy, greater than 65 feet in diameter	141	
	Greater than 1/2 inch	142	
29.	Skirt hem outside stitching margin from edge of webbing		
	1/16 inch up to 1/4 inch		222
	Greater than 1/4 inch	143	
	Any rows of stitching not through skirt band webbing	144	
30.	Tucks		
	1/4 inch wide or less and 1-1/2 inch long or less		223
	Greater than 1/4 inch wide regardless of length	145	
	Greater than 1-1/2 inch long regardless of width	146	
31.	Mismatching of cross seams		
	First seam from vent greater than 2-1/2 inches and 3 inches or less		224

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TABLE XV. Classification of defects – canopies: solid and extended skirt parachute – cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	First seam from vent greater than 3 inches	147	
	All other seams greater than 3 inches and 3-1/2 inches or less		225
	All other seams greater than 3-1/2 inches	148	
32.	Vent lines out of sequence	149	
33.	Suspension and vent line overlap on canopy		
	1/4 inch or less, minus		226
	Greater than 1/4 inch, minus	150	
	Greater than 1 inch, plus		227
34.	Suspension lines not in proper channel or lines crossed	151	
35.	Cut, frayed, or damaged suspension or vent lines	152	
36.	Variation in suspension line length per line group, between vent and skirt		
	1/2 percent or less		228
	Greater than 1/2 percent	153	
37.	Variation in suspension line length per set, between vent and skirt		
	1 percent or less		229
	Greater than 1 percent	154	
38.	Twisted suspension line between vent and skirt (flat)		
	In channel, 360 degrees or less		230
	In channel, greater than 360 degrees	155	
39.	Twisted suspension line between vent and skirt (core or coreless braid)		
	In channel, 720 degrees		231
	In channel, any twist causing kinks in line	156	
40.	Twist between line reinforcement at skirt and 3 inch zigzag stitching above when suspension lines are continuous through canopy (measured by pulling line under 5 pounds of tension between vent and line reinforcement and observing the rotation of zigzag stitching)		
	90 degrees up to 180 degrees		232
	Greater than 180 degrees	157	
	Line not placed between two ply's of material	158	
41.	Variation in suspension line length per line group, between skirt and attaching point		
	1/2 percent or less		233
	Greater than 1/2 percent	159	
42.	Variation in suspension line length per set, between skirt and attaching point		
	1 percent or less		234

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TABLE XV. Classification of defects – canopies: solid and extended skirt parachute – cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Greater than 1 percent	160	
43.	Twisted suspension line between skirt and attaching point (flat), except hem rigged canopy	161	
44.	Twisted suspension line (flat), hem rigged canopy		
	180 degrees up to 360 degrees		235
	Greater than 360 degrees	162	
45.	Twisted suspension line between skirt and attaching point (core and coreless braid)		
	Greater than 360 degrees		236
	Greater than 360 degrees causing kinks in the line	163	
46.	Stitching		
	Stitch per inch defects size 3 - 6 thread:		
	One stitch per inch more or less than the number specified, up to 2 inches in length and one place per seam		237
	One stitch per inch more or less than the number specified, longer than 2 inches in length	164	
	More than one stitch per inch more or less than the number specified	165	
	More than one stitch per inch defect per seam	166	
	Stitch per inch defects size B - FF thread:		
	Two stitches per inch more than the number specified, up to 2 inches in length and one place per seam		238
	Two stitches per inch more or less than the number specified, longer than 2 inches in length	167	
	More than two stitches per inch more or less than the number specified	168	
	More than one stitch per inch defect per seam	169	
	Additional stitch per inch defects, size B and E thread, ASTM D6193 Type 301 or 401 stitch only:		
	More than fourteen stitches per inch	170	
	Fourteen stitches per inch, longer than 2 inches in length	171	
	Fourteen stitches per inch, more than one location per seam, spaced closer than 12 inches apart	172	
	Stitch per inch defects in localized areas of reinforcement or excess thicknesses, classified in this table as "major defect"		239
	NOTE: Stitch per inch defects classified in this table as "minor defect", in localized areas of reinforcement or excess thickness, shall not be classified as a defect.		
	Stitch pattern defects:		
	Broken or missing stitches		
	1/2 inch or less		240
	Greater than 1/2 inch	173	
	Two or more consecutive skipped stitches	174	
	Missing or incomplete stitch pattern (for example, bartack or zigzag)	175	
	Backstitch missing	176	

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TABLE XV. Classification of defects – canopies: solid and extended skirt parachute – cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Backstitch short		
	1/2 inch or less		241
	Greater than 1/2 inch	177	
	Backstitch long, greater than 1/2 inch		242
	Runoff, two or more stitches into single cloth or beyond seam or hem margins		
	1/2 inch or less		243
	Greater than 1/2 inch	178	
	Zigzag stitches with less than 85 percent penetration through suspension lines	179	
	Tension too loose or too tight, 6 inches or less per seam		244
	Tension too loose or too tight, greater than 6 inches per seam	180	
	Kinks, intermittent, spaced closer than 2 inches apart		245
	Stitch pattern lengths not as specified (stitch patterns up to 5 inches long)		246
	1/8 inch or less, minus		
	Greater than 1/8 inch, minus	181	
	Greater than 1/4 inch, plus		247
	Stitch pattern lengths not as specified (stitch patterns over 5 inches long)		248
	1/4 inch or less, minus		
	Greater than 1/4 inch, minus	182	
	Greater than 1/2 inch, plus		249
	Loose or raveled bartack stitching	183	
	Repairs to bartack stitching other than complete removal and replacement	184	
	Defective stitching repaired incorrectly	185	
	Material damaged from stitch rework or repair	186	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	187	
47.	Frequency of repair to stitching (overlaps) on canopies, shall be classified according to the following table		
	More than the number of overlaps in the table	188	

Length of seams (inches)	Overlaps
up to 2	0
greater than 2 up to 12	2
greater than 12 up to 72	4
greater than 72 up to 240	6
Greater than 240	seam length (in) multiplied by .025
Skirt band	seam length (in) multiplied by .02

48.	Length of splices and repairs to stitches		
	Overlap less than 1 inch	189	
	Overlap from 4 inches up to 7 inches		250
	Overlap greater than 7 inches	190	
49.	Pocket band free length		
	1/4 inch or less		251
	Greater than 1/4 inch	191	

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TABLE XV. Classification of defects – canopies: solid and extended skirt parachute – cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
50.	Pocket depth, centerline from main seam to stitch pattern on skirt band		
	1/4 inch or less		252
	Greater than 1/4 inch	192	
51.	Suspension line attaching loop, up to 2 inches in length		
	1/4 inch or less		253
	Greater than 1/4 inch	193	
52.	Suspension line attaching loop, 2 inches to 4 inches in length		
	3/8 inch or less		254
	Greater than 3/8 inch	194	
53.	Location of reefing line cutter bracket from edge of skirt		
	1/4 inch or less		255
	Greater than 1/4 inch	195	
54.	Twisted riser webbing (assembly to links)		
	180 degrees or less, 10 foot web length or less		256
	Greater than 180 degrees, up to 10 foot web length	196	
	180 degrees up to 360 degrees, greater than 10 foot web length		257
	Greater than 360 degrees, greater than 10 foot web length	197	
55.	Riser webbing length		
	1 percent or less		258
	Greater than 1 percent	198	
56.	Damage to textiles, not including cuts, tears and holes		
	1/4 inch long or less and two per canopy 35 feet in diameter or less		259
	Greater than 1/4 inch long or more than two per canopy 35 feet in diameter or less	199	
	1/4 inch long or less and four per canopy 35 feet in diameter and greater		260
	Greater than 1/4 inch long or more than four per canopy 35 feet in diameter and greater	101a	
57.	Cuts, tears or holes		
	Cuts, tears or holes 1/4 inch or less in diameter or 1-1/2 inch or less in length		261
	Cuts, tears or holes greater than 1/4 inch in diameter or 1-1/2 inch in length	102a	
58.	Darns and repairs		
	Darns, more than two per section	103a	
	Darns, more than five per canopy 50 feet in diameter or less	104a	
	Darns, more than ten per canopy from 50 feet up to 75 feet in diameter	105a	
	Darns, more than fifteen per canopy greater than 75 feet in diameter	106a	
	Repair (sewn patch) of a maximum damaged area size of 7 X 7 inches, one per		262

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TABLE XV. Classification of defects – canopies: solid and extended skirt parachute – cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	canopy		
59.	Clear oil stains on canopy material cloth (sewing machine oil)		
	Stains greater than 1/4 square inch (or stains greater than 1/2 inch x 1/2 inch, or 1/4 inch x 1 inch, for example)	107a	
	Two to three stains on a single gore section		263
	More than three stains on a single gore section	108a	
	More than a total of four square inches per entire canopy	109a	
60.	Material unclean, stain (oil, mold, dirt, etc.)	110a	
61.	Identification marking		
	Incorrect or missing	111a	
	Block stamp or printed text illegible		264
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	112a	
62.	Ends of webbing, tape, or cord not seared, waxed, scissor cut or serged or treated with anti-fray compound as specified		265
63.	Searing over stitching	113a	
64.	Thread ends not trimmed		
	1/4 inch up to 2 inches		266
	Greater than 2 inches	114a	

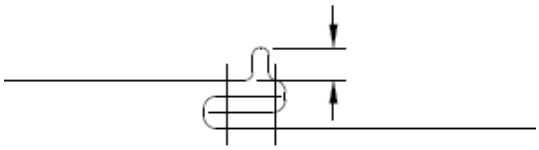
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TABLE XVI. Classification of defects – canopies: cross-cruciform parachute with slider - personnel.

No.	Classification of defects	Classification Major/Critical	Classification Minor
NOTE: T-11 contains center section, arm section, and arm attaching seams.			
NOTE: Single panel section refers to the square sections of cloth that are created in the center section by the grid-like pattern of the center section cross-tapes, and in the arm sections by the arm section radial webbings and horizontal french felled seams.			
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Assembly of grommets		
	Improperly set, two or more exposed teeth for spur washers (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	107	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	108	
	Clinched excessively tight, cutting fabric	109	
	Improper method of installation (for example, base hole in fabric too large)	110	
	Roll of grommet contains splits, burrs or sharp edges, is bent or deformed	111	
8.	Underfold (exposed raw edge)	112	
9.	Overfold and underfold in center section, arm section, arm attaching seams and skirt hem		
	Overfold		
	1/8 inch up to 1/4 inch, any length		201
	Over 1/4 inch, any length	113	
	Underfold		
	1/8 inch up to 1/4 inch, any length		202
	Over 1/4 inch, any length	114	
	1/8 inch up to 1/4 inch, more than 4 inches in length per seam	115	
	More than 3 intermittent underfolds, 1/8 up to 1/4 inch per seam, 2 inches long or less, spaced closer than 12 inches apart	116	
10.	Fullness in seam width, as measured per the following diagram:		

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TABLE XVI. Classification of defects – canopies: cross-cruciform parachute with slider - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
			
	NOTE: T11 center section contains seams in one direction only and no cross seams. Fullness can occur in opposite direction where reinforcement tape is sewn to canopy cloth.		
	Arm attaching or center section seams		
	Greater than 1/8 inch	117	
	Arm section seams or arm vertical tapes		
	Greater than 1/8 inch	118	
11.	Pleats in seams		
	Arm section vertical tapes		
	1/16 inch up to 1/8 inch and one per seam		203
	1/16 inch up to 1/8 inch and more than one per seam	119	
	1/16 inch up to 1/8 inch and more than six per canopy	120	
	Greater than 1/8 inch	121	
	Arm section and center section seams		
	1/16 inch up to 1/8 inch and one per seam		204
	1/16 inch up to 1/8 inch and more than one per seam	122	
	1/16 inch up to 1/8 inch and more than six per canopy	123	
	Greater than 1/8 inch	124	
	Center section seams connecting reinforcement panels		
	1/8 inch up to 3/16 inch and one per corner		205
	1/8 inch up to 3/16 inch and more than one per corner	125	
	Greater than 3/16 inch	126	
12.	Double box-x (at corners of center section) short legs not adjacent to corner vent opening	127	
13.	Misalignment of reinforcement tapes		
	Arm section vertical tape not fully within the width of center section reinforcement tape	128	
	Arm section vertical tape and center section reinforcement tape not fully captured by box-x	129	
14.	Skirt hems improperly formed		
	Raw edge showing	130	
	Intermittent underfold, 50 percent of seam width or greater, where edge is caught with one row of stitching		206
	Underfold where edge is caught with one row of stitching		
	1/4 inch or less, any length		207
	Greater than 1/4 inch, any length	131	

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TABLE XVI. Classification of defects – canopies: cross-cruciform parachute with slider - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Skirt hem formed by more than two pieces of webbing or tape	132	
15.	Wrinkles, pleats, or folds in skirt hem that do not extend into the arm section		
	1/8 inch or less and one per gore		208
	1/8 inch or less, and less than four per canopy		209
	Greater than 1/8 inch	133	
16.	Wrinkles, pleats, or folds in skirt hem extending into the arm section		
	1/8 inch or less, and one per gore		210
	1/8 inch or less, and less than four per canopy		211
	Greater than 1/8 inch	134	
17.	Skirt hem outside stitching margin from edge of webbing		
	1/16 inch up to 3/16 inch		212
	Greater than 3/16 inch	135	
	Any row of stitching not through skirt band webbing	136	
18.	Tucks, where section of fabric caught in stitching of seams or hems	137	
19.	Cut, frayed, or damaged suspension lines	138	
20.	Twisted suspension line between skirt and connector link		
	Greater than 90 degrees up to 180 degrees		213
	Greater than 180 degrees	139	
21.	Open seams	140	
22.	Stitching		
	Stitch per inch defects:		
	Two stitches per inch more or less than the number specified, up to 2 inches in length, and one place per seam		214
	More than two stitches per inch more or less than the number specified, longer than 2 inches in length	141	
	More than two stitches per inch more or less than the number specified	142	
	More than one stitch per inch defect per seam	143	
	Additional stitch per inch defects, size B and E thread, ASTM D6193 Type 301 or 401 stitch only:		
	More than fourteen stitches per inch	144	
	Fourteen stitches per inch, longer than 2 inches in length	145	
	Fourteen stitches per inch, more than one location per seam, spaced closer than 12 inches apart	146	
	Stitch pattern defects:		
	Any broken or missing stitch	147	
	Two or more consecutive skipped stitches	148	
	Missing or incomplete stitch pattern, for example, 3-point, box-x, bartack,	149	

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TABLE XVI. Classification of defects – canopies: cross-cruciform parachute with slider - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	zigzag		
	Backstitch missing	150	
	Backstitch short		
	Greater than 1/16 inch up to 1/8 inch		215
	Greater than 1/8 inch	151	
	Backstitch long, greater than 1/4 inch		216
	Runoff, two or more stitches into single cloth or beyond seam or hem margins		
	1/4 inch or less		217
	Greater than 1/4 inch	152	
	Tension too loose or too tight		
	3 inches long or less, one per seam		218
	More than 3 inches long per seam	153	
	More than one area per seam	154	
	Kinks, intermittent, spaced closer than 2 inches apart		219
	Stitch pattern lengths not as specified (stitch patterns 5 inches long or less)		
	Greater than 1/8 inch, minus	155	
	Greater than 1/8 inch, plus		220
	Stitch pattern lengths not as specified (stitch patterns greater than 5 inches long)		
	1/4 inch or less, plus or minus		221
	Greater than 1/4 inch, minus	156	
	Greater than 1/4 inch, plus		222
	Bartack stitch pattern length not as specified in TDP	157	
	Box-x stitch pattern length not as specified in TDP	158	
	Defective stitching not repaired correctly or exceeding specified limits	159	
	Defective stitching repaired when repair is not authorized	160	
	Any damage to material resulting from stitch rework or repair	161	
	Frequency of repairs to stitching exceeding specified limits	162	
	Loose or raveled bartack stitching	163	
	Repairs to bartack stitching other than complete removal and replacement	164	
	Resewn stitch patterns and bartacks exceeding specified limits	165	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	166	
23. Stitch overlaps			
	NOTE: One stitch repair or splice will result in two overlaps, one overlap on either end of the original stitch, regardless of number of needles used (single or multi-needle machine).		
	Center section main seams, center section vertical tapes, arm attaching seams	167	
	More than eight stitch overlaps per seam		
	Arm section seams	168	
	More than seven stitch overlaps per seam		
	Arm section radial webbing	169	
	More than five stitch overlaps per section		
	Skirtband	170	
	More than five stitch overlaps per entire skirtband		

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TABLE XVI. Classification of defects – canopies: cross-cruciform parachute with slider - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
24.	Frequency of stitch overlaps (from stitch repairs or splices) that are not otherwise addressed in this defect table shall be classified according to the following table		
	More than the number of overlaps in the following table	171	
	NOTE: One stitch repair or splice will result in two overlaps, one overlap on either end of the original stitch, regardless of number of needles used (single or multi-needle machine).		

Length of seams (inches)	Overlaps
2 or less	0
Greater than 2 up to 12	2
Greater than 12 up to 72	4
Greater than 72 up to 240	6
Greater than 240	seam length (in) multiplied by .025

25.	Length of stitch overlaps		
	Less than 2 inches	172	
	Greater than 4 inches		223
26.	Damage to textiles, not including cuts, tears or holes		
	1/8 inch long or less and one per canopy		224
	Greater than 1/8 inch long	173	
	More than one per canopy	174	
27.	Cuts, tears and holes (except necessary drill holes) to canopy cloth		
	Damage resulting in cuts, tears, or holes involving three or more warp or filling yarns	175	
28.	Cuts, tears, darns and repairs on mesh panels	176	
29.	Cuts, tears, darns and repairs to tape or webbing	177	
30.	Darns on canopy cloth		
	More than one per section	178	
	More than three per canopy	179	
31.	Splicing of any cord, webbing or tape unless otherwise specified	180	
32.	Oil stains on canopy material		
	Stains greater than 1/4 square inch (or stains greater than 1/2 inch x 1/2 inch, or 1/4 inch x 1 inch, for example)	181	
	Two to three stains on a single panel section		225
	More than three stains on a single panel section	182	
	More than a total of 4 square inches per entire canopy	183	
33.	Material unclean, stain (oil, mold, dirt, etc.)	184	

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TABLE XVI. Classification of defects – canopies: cross-cruciform parachute with slider - personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
34.	Identification marking		
	Incorrect or missing	185	
	Block stamp or printed text illegible		226
	Actual part number, contract number, serial number, DOM or contractor CAGE code illegible	186	
35.	Ends of webbing, tape, or cord not seared, waxed, scissor cut or serged or treated with anti-fray compound as specified		227
36.	Searing over stitching	187	
37.	Thread ends not trimmed		
	1/4 inch up to 2 inches		228
	Greater than 2 inches	188	

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TABLE XVII. Classification of defects – deployment sleeve – personnel.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Seam folding		
	Improperly formed seam or hem exposing raveled edge of fabric	107	
	Stow panel not captured by all rows of stitching and bartack	108	
	Bridle loop not captured by all rows of stitching and bartack	109	
	Drogue loop not captured by all rows of stitching and bartack	110	
8.	Splicing of cloth, tape or webbing	111	
9.	Stitching		
	Up to two stitches per inch more or less than the number specified, up to 2 inches in length		201
	Up to two stitches per inch more or less than the number specified, longer than 2 inches in length	112	
	More than two stitches per inch more or less than the number specified	113	
	Any broken or missing stitch	114	
	Two or more consecutive skipped stitches	115	
	Runoff, over 1/2 inch in length in any one place or greater than 1-1/2 inch total length per sleeve	116	
	Missing or incomplete stitch pattern (for example, bartack or cross box)	117	
	Backstitch missing	118	
	Backstitch short		
	1/4 inch or less		202
	Greater than 1/4 inch	119	
	Backstitch long, greater than 1/4 inch		203
	Tension too loose or too tight		
	4 inches or less per seam		204
	Greater than 4 inches per seam	120	
	Kinks, intermittent, spaced closer than 2 inches apart		205
	Stitch pattern lengths not as specified (stitch patterns 5 inches long or less)		
	Bartack stitch pattern length greater than 1/8 inch, plus		206
	Repairs to bartack stitching other than complete removal and replacement	121	
	Loose or raveled bartack stitching	122	
	Defective stitching repaired incorrectly, or exceeding specified limits	123	

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TABLE XVII. Classification of defects – deployment sleeve – personnel - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Any damage to material resulting from stitch rework or repair	124	
	Bird nest (stitching nest) including masses of knotted and tangled thread	125	
10.	Damage to textiles, not including cuts, tears and holes	126	
11.	Cuts, tears and holes (except necessary drill holes)	127	
12.	Any darn	128	
13.	Pleats greater than 1/16 inch		207
14.	Edge of hook tape extends beyond folded edge of cotton	129	
15.	Edge of loop tape extends beyond folded edge of cotton		208
16.	Identification marking		
	Incorrect or missing	130	
	Block stamp or printed text illegible		209
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	131	
17.	Ends of webbing, tape, or cord not seared, waxed, scissor cut, serged or treated with anti-fray compound as specified		210
18.	Searing over stitching	132	
19.	Material unclean, stain (oil, mold, etc.)	133	
20.	Thread ends not trimmed		
	1/4 inch up to 2 inches		211
	Greater than 2 inches	134	

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TABLE XVIII. Classification of defects – canopies: square, gliding type - cargo.

No.	Classification of defects	Classification Major/Critical	Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Improper cascade continuity and suspension line routing continuity, one or more lines	107	
8.	Canopy out of trim more than allowed per TDP	108	
9.	Canopy bottom skin not in accordance with TDP	109	
10.	Assembly of grommets		
	Improperly set, two or more exposed teeth (except where grommet is set as specified but is in an area of unequal thickness that prevents uniform clinching)	110	
	Clinched loosely, allowing grommet to rotate in hole for spur washers or where grommet/washer may become disengaged	111	
	Clinched excessively tight, causing cut fabric	112	
	Improper method of installation (for example, base hole in fabric too large)	113	
	Roll of grommet contains splits, burrs or sharp edges, is bent or deformed	114	
11.	Any seam not constructed as specified	115	
12.	Open seam	116	
13.	Cut, frayed, or damaged suspension line	117	
14.	Twisted suspension line between canopy and junction, greater than 360 degrees	118	
15.	Twisted suspension line between junction and connector link, greater than 360 degrees	119	
16.	Twisted control line between canopy trailing edge and upper cascade, greater than 360 degrees	120	
17.	A, B, C and D reinforcement tape on ribs, take-up or stretch greater than 3/4 inch	121	
18.	Rib leading edge tape		

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TABLE XVIII. Classification of defects – canopies: square, gliding type - cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
	Any one rib greater than 1/2 inch, plus or minus	122	
	More than eight ribs per canopy out of tolerance	123	
	NOTE: Up to eight ribs up to 1/2 inch, plus or minus, will not be counted as a defect		
19.	Reinforcement tape and suspension line tape, rib panel construction		
	Missing	124	
	Improperly attached	125	
	NOTE: The bartack used to secure the reinforcement tape within the seam shall not cross the primary row of double needle seam stitching. The primary row of stitching shall be understood as the immediate stressed row when load is placed on the seam.		
20.	Cloth not heat seared as specified in the TDP	126	
21.	Heat sear of cross port vent holes improperly formed		
	Three per rib		201
	More than twelve per canopy	127	
22.	Splicing		
	On canopy fabric	128	
	Webbing or tape, unless splice is detailed in TDP		202
23.	Stitching		
	301 and 308 stitching not in accordance with ASTM D6193, more than one stitch per inch more or less than the number specified		
	2 inches or less		203
	Greater than 2 inches	129	
	Backstitch missing	130	
	Incorrect backstitch	131	
	Backstitch short		
	Greater than 1/16 inch up to 1/8 inch		204
	Greater than 1/8 inch	132	
	Backstitch long, greater than 1/4 inch		205
	Stitch pattern length not in accordance with ASTM D6193 or applicable drawing		
	1/4 inch or less, minus		206
	Greater than 1/4 inch, minus	133	
	Greater than 1/2 inch, plus		207
	Missing or incomplete stitch pattern (for example, bartack or cross box)	134	
	Bartack stitch pattern not as specified, canopy only		
	Length, greater than 1/16 inch, plus	135	
	Width, greater than 1/32 inch, plus	136	
	Bartack stitch pattern not as specified, suspension and steering lines only		
	Length greater than 1/16 inch, plus	137	

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TABLE XVIII. Classification of defects – canopies: square, gliding type - cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Skips in bartack stitching, line attachment bartacks only		
	More than two skips in any bartack	138	
	Three bartacks on a single line attachment containing skips		208
	Repairs to bartack stitching other than complete removal and replacement	139	
	Loose or raveled bartack stitching	140	
	Two or more consecutive broken, missed, or skipped stitches	141	
	Runoff, two or more stitches into single cloth or beyond seam or hem margins		
	1/2 inch or less		209
	Greater than 1/2 inch	142	
	Tension too loose or too tight		
	6 inches or less per seam		210
	Greater than 6 inches per seam	143	
	Kinks, intermittent, spaced closer than 2 inches apart		211
	Defective stitching repaired incorrectly	144	
	Any damage to material resulting from stitch rework or repair	145	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	146	
24.	Damage to textiles not including holes, cuts or tears		
	1/4 inch or less in length and two per canopy		212
	Greater than 1/4 inch long or more than two per canopy	147	
25.	Holes (except necessary drill holes), cuts and tears	148	
26.	Darns		
	More than one per cell	149	
	More than six per canopy	150	
27.	Ends specified as scissor cut but are heat seared	151	
28.	Ends of webbing, tape and cord not seared, waxed, scissor cut or serged as specified		213
29.	Identification marking		
	Incorrect or missing	152	
	Block stamp or printed text illegible		214
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	153	
30.	Searing over stitching	154	
31.	Thread ends not clipped		
	1/4 inch up to 2 inches		215
	Greater than 2 inches	155	

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TABLE XIX. Classification of defects – harness: joint precision airdrop system - cargo.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Twisted or malformed part	107	
8.	Stitching		
	301 stitching not in accordance with ASTM D6193		
	One stitch per inch more or less than the number specified		201
	More than one stitch per inch more or less than the number specified	108	
	Stitch pattern length not in accordance with applicable drawings		
	Stitch patterns 5 inches long or less		
	1/8 inch or less, plus or minus		202
	Greater than 1/8 inch, minus	109	
	Greater than 1/8 inch, plus		203
	Stitch patterns greater than 5 inches long		
	1/4 inch or less, plus or minus		204
	Greater than 1/4 inch, minus	110	
	Greater than 1/4 inch, plus		205
	Stitch pattern width not in accordance with applicable drawing		
	Greater than 1/16 inch, plus	111	
	Missing or incomplete stitch pattern	112	
	Backstitch missing	113	
	Backstitch short		
	1/4 inch or less		206
	Greater than 1/4 inch	114	
	Backstitch long, greater than 1/4 inch		207
	Broken, missing, or skipped stitches 1/2 inch or greater in length	115	
	Tension too loose or too tight in any one row		
	4 inches or less		208
	Greater than 4 inches	116	
	Loops in stitching due to varying thicknesses of webbings		
	1/4 inch or less		209
	Greater than 1/4 inch	117	
	Loops, spaced closer than 2 inches apart		210
	Kinks, intermittent, spaced closer than 2 inches apart		211
	Defective stitching not repaired correctly	118	
	Repairs to bartack stitching other than complete removal and replacement	119	
	Loose or raveled bartack stitching	120	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	121	

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TABLE XIX. Classification of defects – harness: joint precision airdrop system - cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	Runoff, greater than 1/2 inch in any one place	122	
	Damage to material resulting from stitch rework or repair	123	
9.	Material unclean, stain (oil, mold, dirt, etc.)	124	
10.	Identification marking		
	Incorrect or missing	125	
	Block stamp or printed text illegible		212
	Actual part number, contract number, serial number, DOM, or contractor CAGE code illegible	126	
11.	Ends of webbing, tape, or cord not seared, waxed, scissor cut or serged as specified		213
12.	Searing over stitching	127	
13.	Thread ends not clipped		
	1/4 inch up to 2 inches		214
	Greater than 2 inches	128	

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TABLE XX. Classification of defects – soft link - cargo.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Material with major or critical defects, as specified in the material specifications	105	
6.	Any foreign object found	106	
7.	Missing or incomplete bartack	107	
8.	Bartack stitch pattern length not as specified		
	Length greater than 1/16 inch, plus	108	
	Width greater than 1/32 inch, plus	109	
9.	Repairs to bartack stitching other than complete removal and replacement	110	
10.	Damage to textile (cuts, chews, holes, etc), any	111	
11.	Thread ends not trimmed		
	1/4 inch up to 2 inches		201
	Greater than 2 inches	112	

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TABLE XXI. Classification of defects – parachute system: one time use - cargo.

No.	Classification of defects	Classification Major/Critical	Classification Minor
1.	Any dimension out of tolerance per TDP unless otherwise specified in this table	101	
2.	Improper assembly (construction detail, note, etc.) per TDP	102	
3.	Component missing per TDP	103	
4.	Material not as specified per TDP	104	
5.	Any foreign object found	105	
6.	Loose, broken, or sharp edges on grommets	106	
7.	Stitching		
	Stitch per inch defects, size 3 - 6 thread:		
	One stitch more or less than the number specified		
	Up to 2 inches long, and one place per seam		201
	Greater than 2 inches long	107	
	More than one stitch more or less than the number of specified	108	
	Stitch per inch defects, size B - FF thread:		
	Two stitches more or less than the number specified		
	Up to 2 inches long and one place per seam		202
	Greater than 2 inches long	109	
	More than two stitches more or less than the number specified	110	
	Additional stitch per inch defects, size B and E thread, ASTM D6193 Type 301 or 401 stitch only:		
	More than fourteen stitches per inch	111	
	Fourteen stitches per inch, longer than 2 inches in length	112	
	Fourteen stitches per inch, two or more locations per seam, spaced closer than 12 inches apart	113	
	Stitch per inch defects; risers, static line, bridle, strap, adaptor web, handle:		
	Two stitches more or less than the number specified, up to 4 inches in length		203
	Two stitches more or less than the number specified, longer than 4 inches in length	114	
	More than two stitches more or less than the number specified	115	
	Stitch per inch defects in localized areas of reinforcement or excess thickness, classified in this table as "major defect"		204
	NOTE: Stitch per inch defects classified in this table as "minor defect", in localized areas of reinforcement or excess thickness, shall not be classified as a defect.		
	Broken, missing or skipped stitches		
	Two or more consecutive	116	
	Greater than 1/2 inch in length in any one place per deployment bag, riser, static line, bridle, strap	117	
	Greater than 1-1/2 inch total length per deployment bag, riser, static line, bridle, strap	118	
	Missing or incomplete stitch pattern (for example, bartack or zigzag)	119	
	Incorrect backstitch	120	
	Backstitch missing	121	
	Backstitch short		

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TABLE XXI. Classification of defects – parachute system: one time use - cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
	1/4 inch or less		205
	Greater than 1/4 inch	122	
	Backstitch long, greater than 1/4 inch		206
	Bartack stitch pattern not as specified, canopy only		
	Length greater than 1/16, plus	123	
	Width greater than 1/32, plus	124	
	Bartack stitch pattern not as specified, suspension and steering lines only		
	Length greater than 1/16, plus	125	
	Skips in bartack stitching, line attachment bartacks only		
	More than two skips in any bartack	126	
	Repairs to bartack stitching other than complete removal and replacement	127	
	Loose or raveled bartack stitching	128	
	Runoff, two or more stitches into single cloth or beyond seam or hem margins		
	1/2 inch or less		207
	Greater than 1/2 inch	129	
	Tension too loose or too tight		
	6 inches or less per parachute or slider seam		208
	Greater than 6 inches per parachute or slider seam	130	
	4 inches or less per pilot chute and deployment bag seam		209
	Greater than 4 inches per pilot chute and deployment bag seam	131	
	Kinks, intermittent, spaced closer than 2 inches apart		210
	Stitch loop in riser, static line, bridle, strap due to sewing over various layers of webbing		
	1/8 inch up to 1/4 inch		211
	Greater than 1/4 inch	132	
	Stitch pattern length not in accordance with ASTM D6193 or applicable drawing		
	Stitch patterns 5 inches or less in length		
	1/8 inch or less, minus		212
	Greater than 1/8 inch, minus	133	
	Greater than 1/4 inch, plus		213
	Stitch patterns greater than 5 inches long		
	1/4 inch or less, minus		214
	Greater than 1/4 inch, minus	134	
	Greater than 1/2 inch, plus		215
	Defective stitching repaired incorrectly	135	
	Damage to material resulting from stitch rework or repair	136	
	Bird nest (stitching nest) to include masses of knotted and tangled thread	137	
8.	Identification marking illegible, incorrect or missing		216
9.	Ends of webbing, tape, or cord not seared, waxed, scissor cut or serged as specified		217
10.	Searing over stitching	138	
11.	Untrimmed thread ends greater than 2 inch in length	139	

PARACHUTE AND SLIDER

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TABLE XXI. Classification of defects – parachute system: one time use - cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Classification Minor
12.	Improper type or improper assembly of cross and main seams	140	
13.	Suspension and control lines misrouted	141	
14.	Trim out of tolerance	142	
15.	Improper cascade continuity	143	
16.	Twisted suspension or control line		
	180 degrees up to 360 degrees		218
	Greater than 360 degrees	144	
17.	Snagged suspension or control line		
	One carrier, 1/4 inch or less		219
	More than one carrier or greater than 1/4 inch	145	
18.	Frayed suspension or control lines resulting in 1/4 of the fibers within one carrier being cut	146	
19.	Any seam not constructed as specified	147	
20.	More than five pleats per seam, greater than 1/8 inch wide	148	
21.	Rib leading edge tape		
	Any one rib greater than 1/2 inch, plus or minus	149	
	More than eight ribs per canopy out of tolerance	150	
	NOTE: Eight ribs or less, 1/2 inch or less, plus or minus will not be counted as a defect.		
22.	Damage to textiles, not including cuts, tears and holes		
	1/4 inch long or less and four per canopy		220
	Greater than 1/4 inch long or more than four per canopy	151	
23.	Cuts, tears or holes		
	1/4 inch or less in diameter or 1-1/2 inch or less in length		221
	Greater than 1/4 inch in diameter or 1-1/2 inch in length	152	
24.	Darns and repairs		
	Darns, more than two per panel	153	
	Darns, more than five per canopy 50 feet in diameter or less	154	
	Sewn patch, more than two per canopy	155	
	PILOT CHUTE		
25.	Radial seam length		
	1/4 inch or less		222
	Greater than 1/4 inch	156	

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TABLE XXI. Classification of defects – parachute system: one time use - cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
26.	Gore width at skirt, centerline to centerline on radial seam		
	1/8 inch or less		223
	Greater than 1/8 inch	157	
27.	Skirt webbing overlap		
	1/4 inch or less, minus		224
	Greater than 1/4 inch, minus	158	
	Greater than 1/4 inch, plus		225
28.	Seam folding, raw edge	159	
29.	Pleats		
	1/16 inch up to 1/8 inch and one per gore		226
	Greater than 1/8 inch or more than one per gore	160	
30.	Loop length		
	1/8 inch or less, minus		227
	Greater than 1/8 inch, minus	161	
	Greater than 1/4 inch, minus		228
	DEPLOYMENT BAG, CONTAINER		
31.	Seam, improperly formed	162	
32.	Finished dimensions, overall outside, including bag circumference		
	1/4 inch or less		229
	Greater than 1/4 inch	163	
33.	Grommet spacing		
	1/8 inch or less		230
	Greater than 1/8 inch	164	
34.	Pleats greater than 1/16 inch	165	
35.	Damage to textiles, not including cuts, tears and holes		
	1/4 inch or less and two per bag		231
	Greater than 1/4 inch or more than two per bag	166	
36.	Holes (except necessary drill holes), darns and repairs		
	Holes greater than 1/8 inch diameter	167	
	Holes less than 1/16 inch diameter, three or less per bag		232
	Holes, more than three per bag, any size	168	
	Darns and repairs		
	1/4 inch up to 1/2 inch		233
	Greater than 1/2 inch	169	
	Darns, more than four per bag, any size	170	
	More than one patch per bag	171	

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TABLE XXI. Classification of defects – parachute system: one time use - cargo - Continued.

No.	Classification of defects	Classification Major/Critical	Minor
	RISERS, STATIC LINES, BRIDLES, STRAPS		
37.	Length (finished dimensions)		
	1 percent or less		234
	Greater than 1 percent	172	
38.	Twisted or malformed part	173	
39.	Spacing dimensions		
	1 percent or less		235
	Greater than 1 percent	174	
40.	Loop, 2 inches or less in length		
	1/8 inch or less		236
	Greater than 1/8 inch	175	
41.	Loop, greater than 2 inches in length		
	1/4 inch or less		237
	Greater than 1/4 inch	176	

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CONCLUDING MATERIAL

Custodians:
Army - GL
Navy - AS
Air Force - 11

Preparing activity:
DLA - GS1
(Project 1670-2019-002)

Review activities:
Army – AV, TE
Navy - 71

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